

ML Hybrid Models Comparison Report

Generated: 2025-12-05T17:53:28.634533

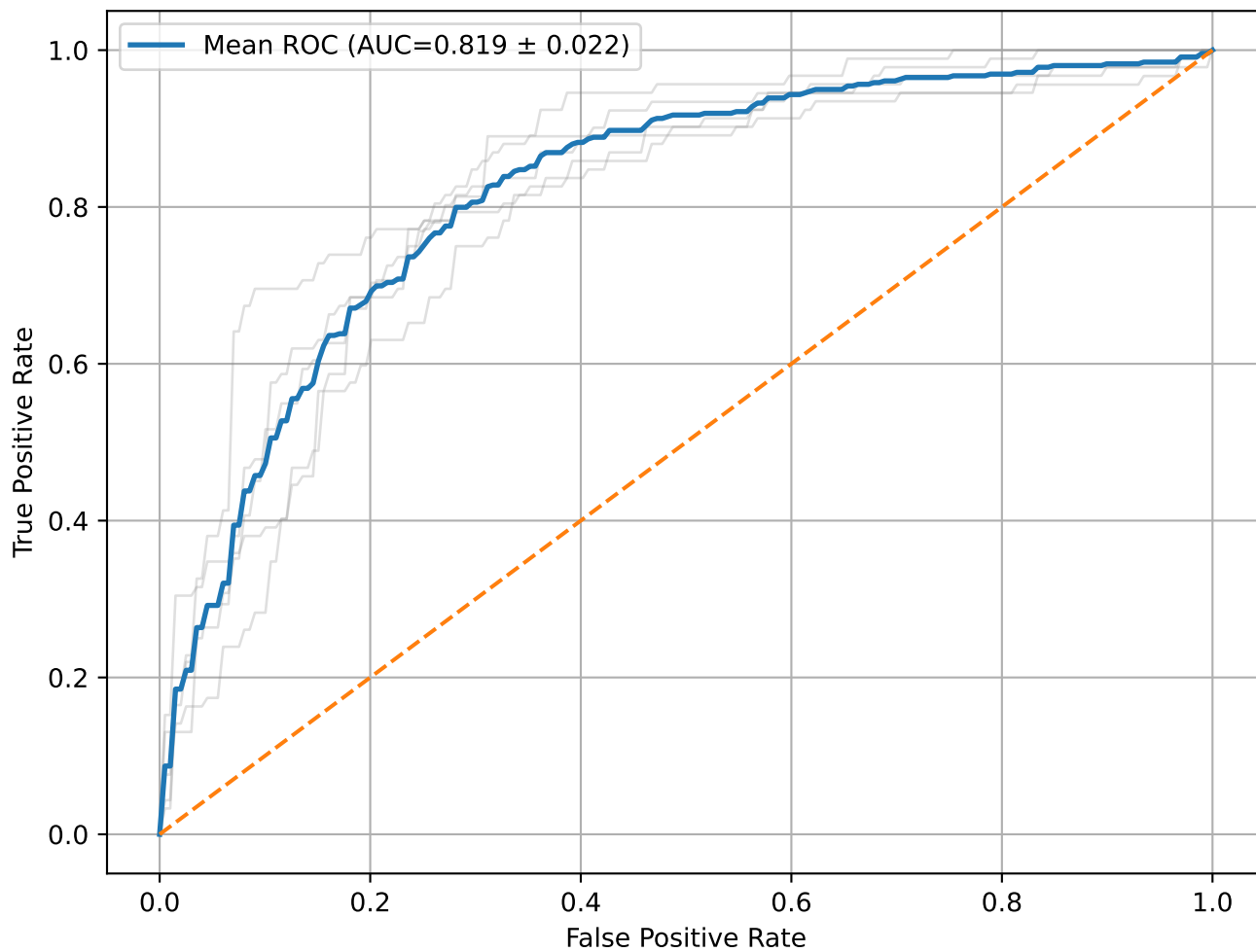
Dataset samples: 906

Models: logistic, rf, svm, xgb, lgb, dnn, stacking, voting, bagging, pca_svm, ga_xgb, pso_svm, bayes_opt_svms

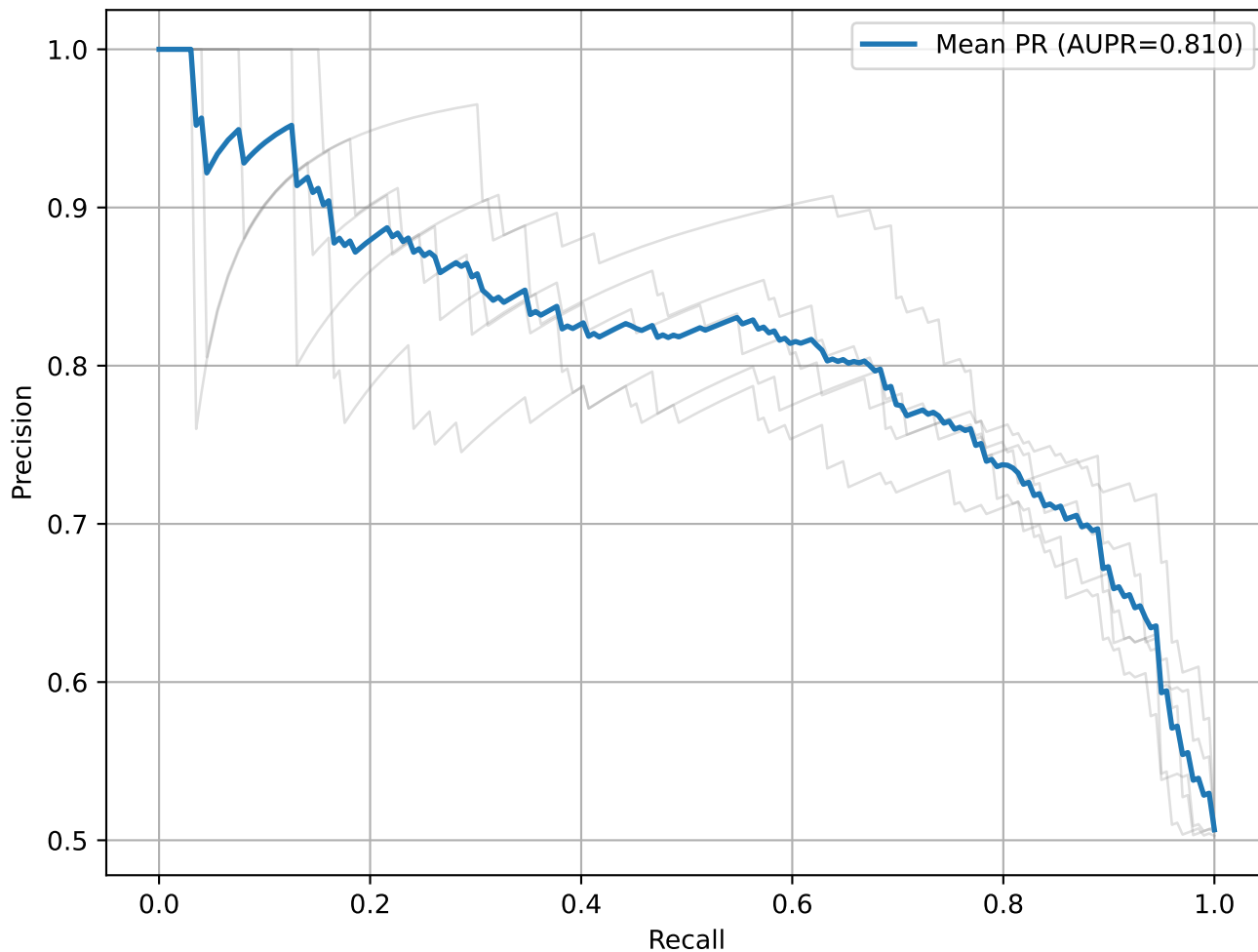
StratifiedKFold: 5 splits

Random seed: 475

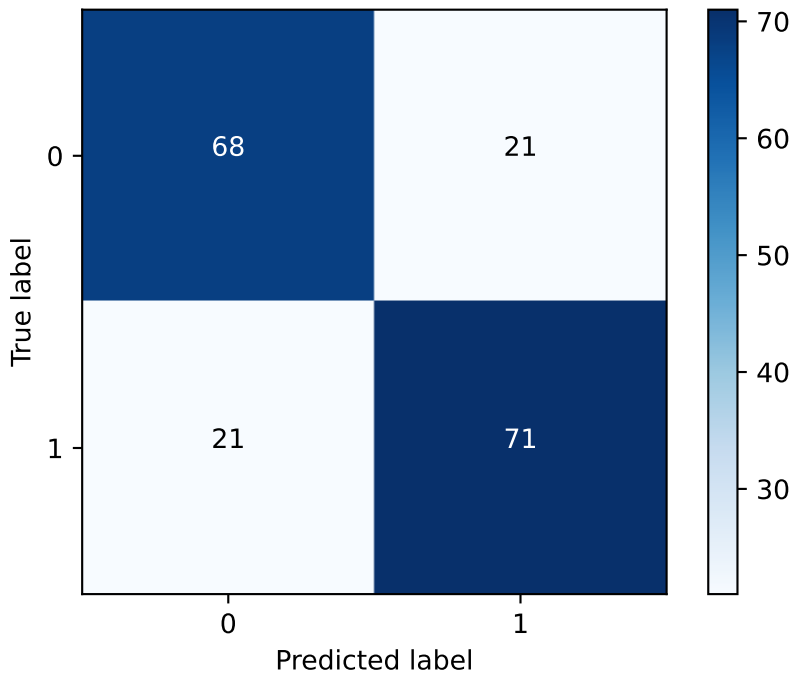
ROC - LOGISTIC



Precision-Recall - LOGISTIC



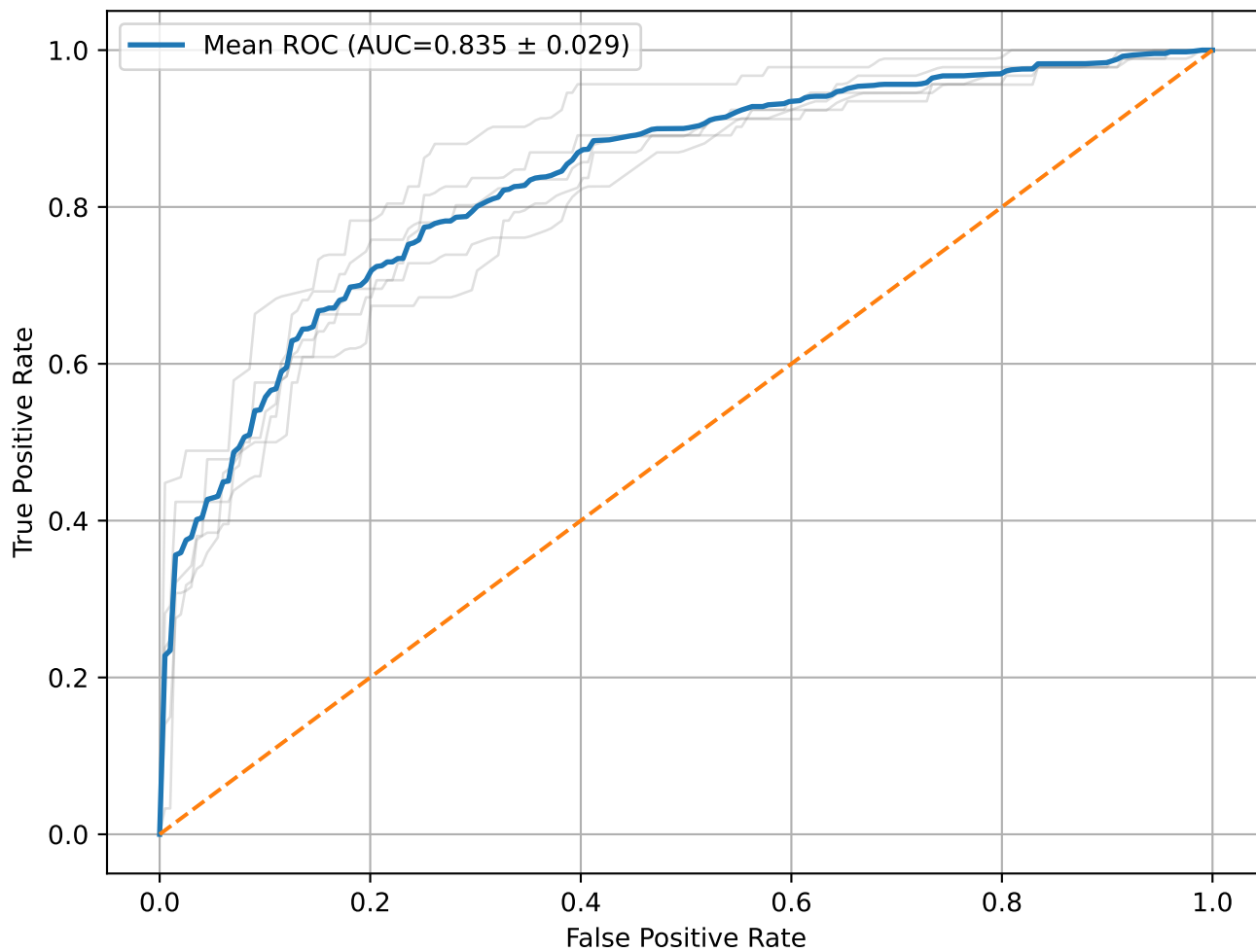
Confusion Matrix - LOGISTIC (last fold)



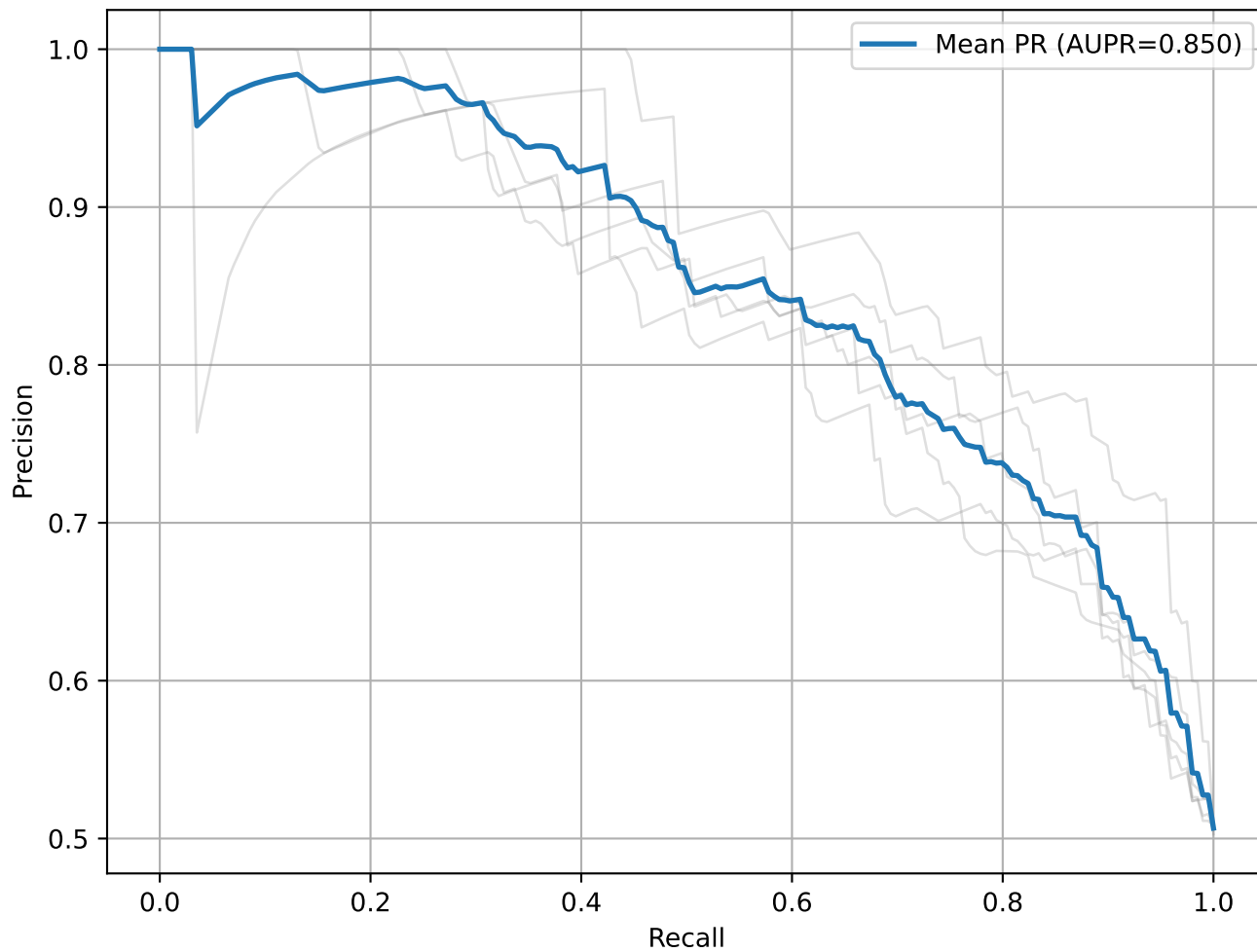
Classification Report - LOGISTIC (last fold)

	precision	recall	f1-score	support
0	0.76	0.76	0.76	89
1	0.77	0.77	0.77	92
accuracy			0.77	181
macro avg	0.77	0.77	0.77	181
weighted avg	0.77	0.77	0.77	181

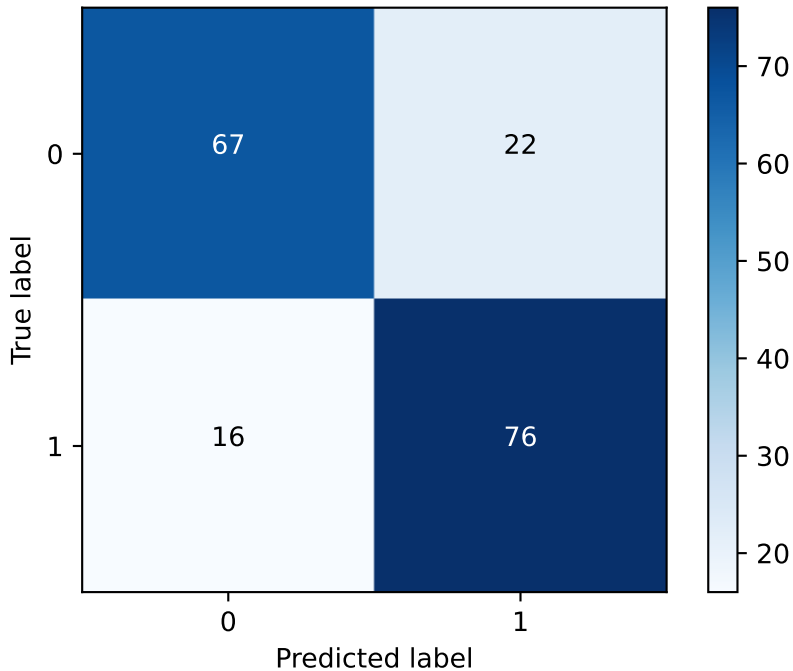
ROC - RF



Precision-Recall - RF



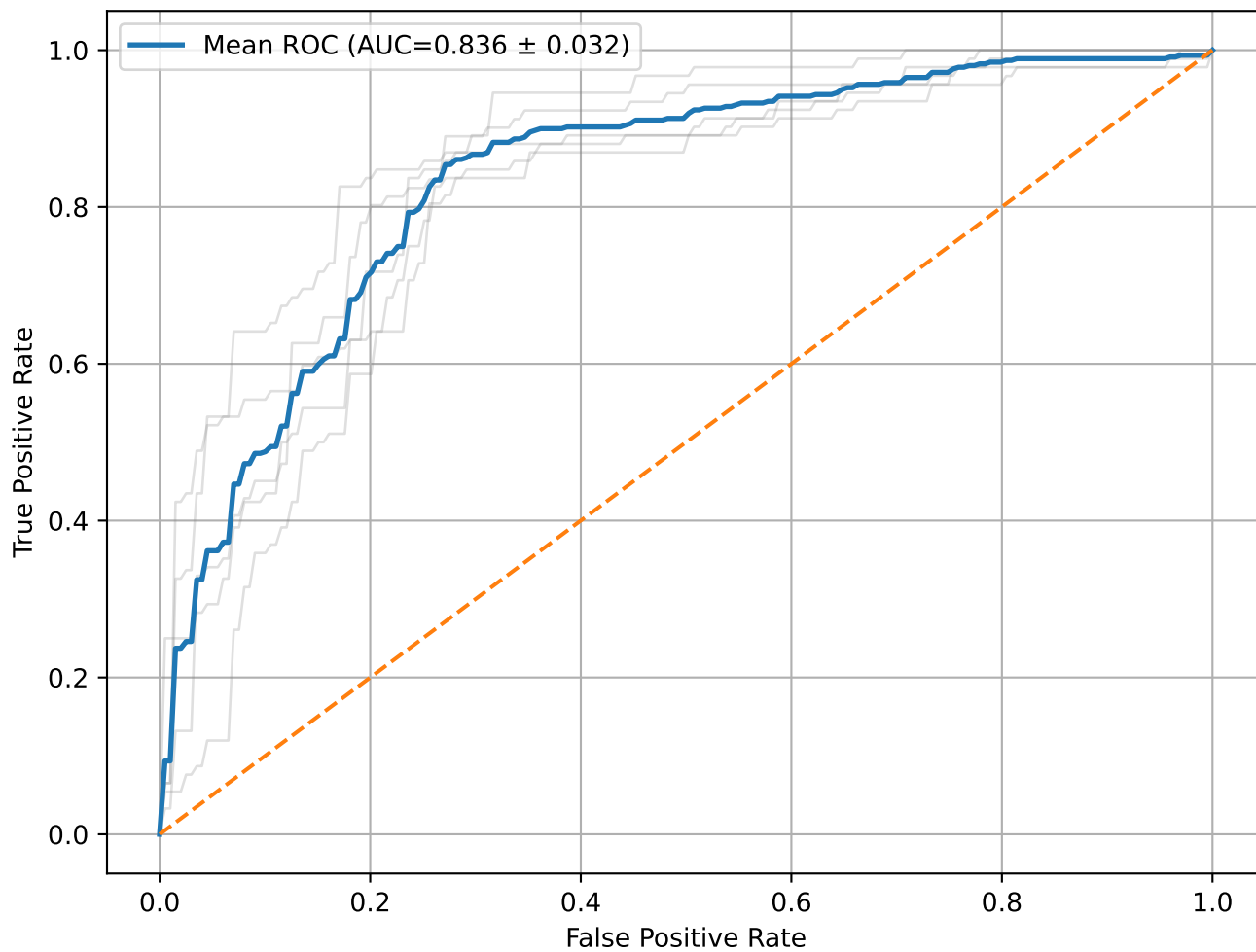
Confusion Matrix - RF (last fold)



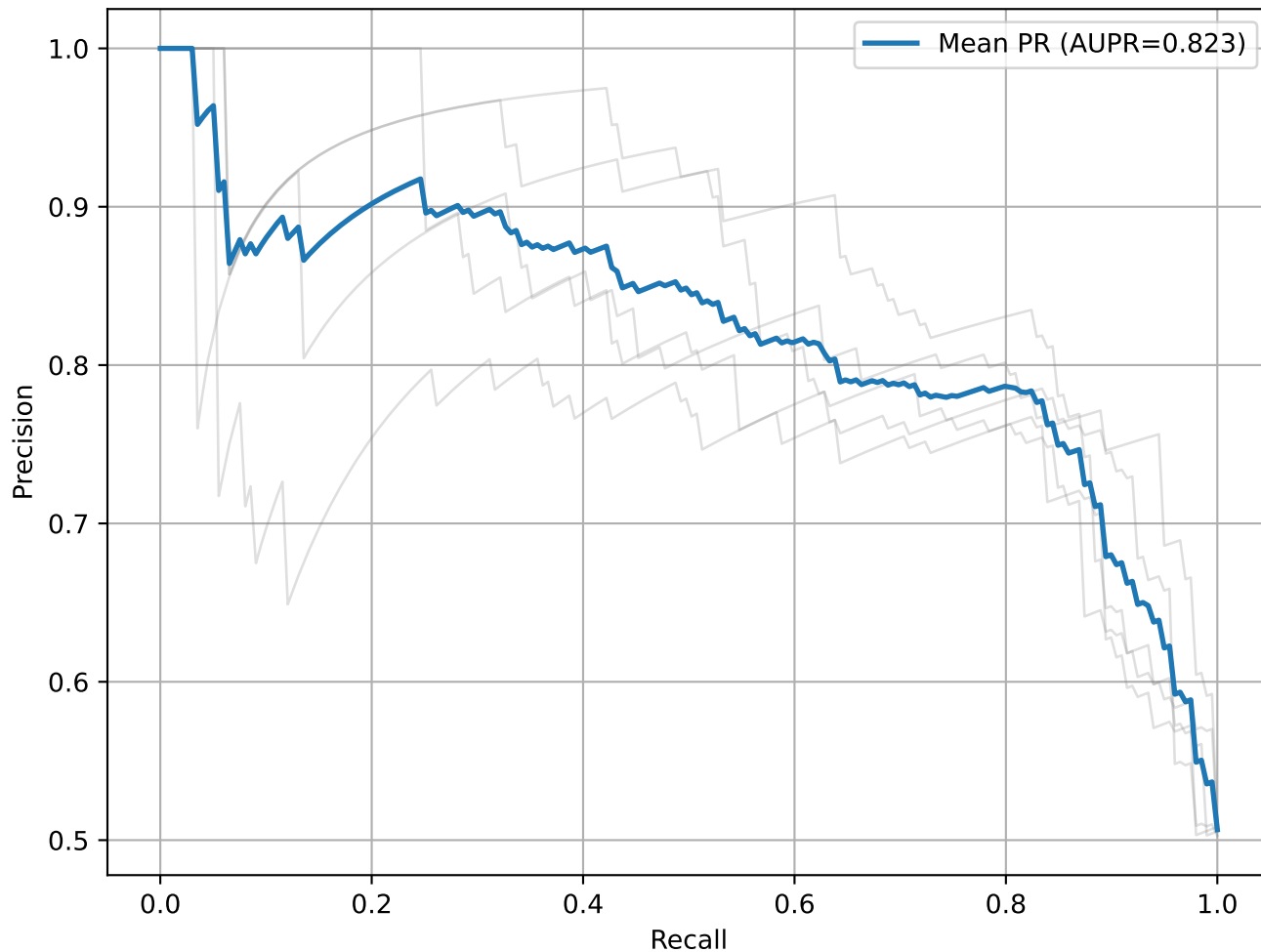
Classification Report - RF (last fold)

	precision	recall	f1-score	support
0	0.81	0.75	0.78	89
1	0.78	0.83	0.80	92
accuracy			0.79	181
macro avg	0.79	0.79	0.79	181
weighted avg	0.79	0.79	0.79	181

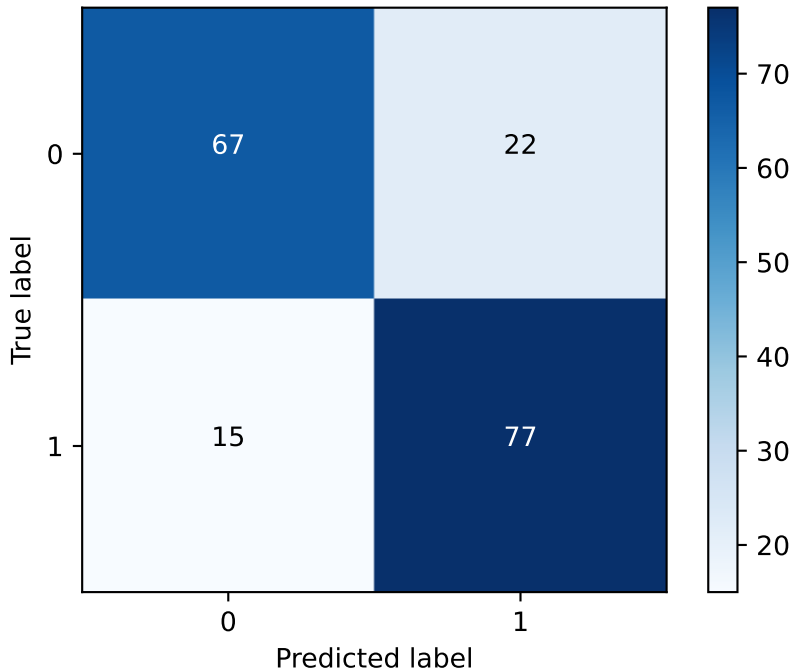
ROC - SVM



Precision-Recall - SVM



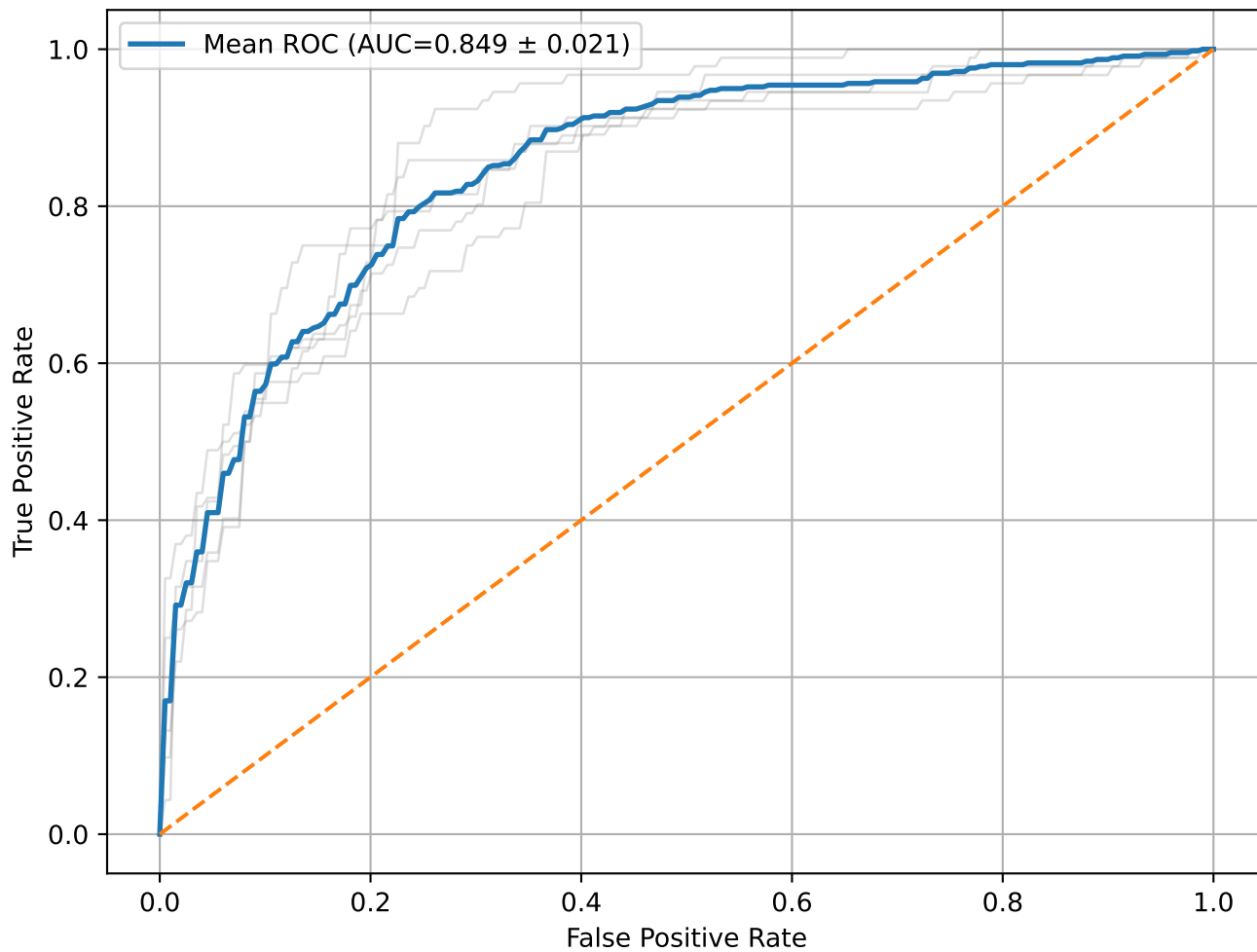
Confusion Matrix - SVM (last fold)



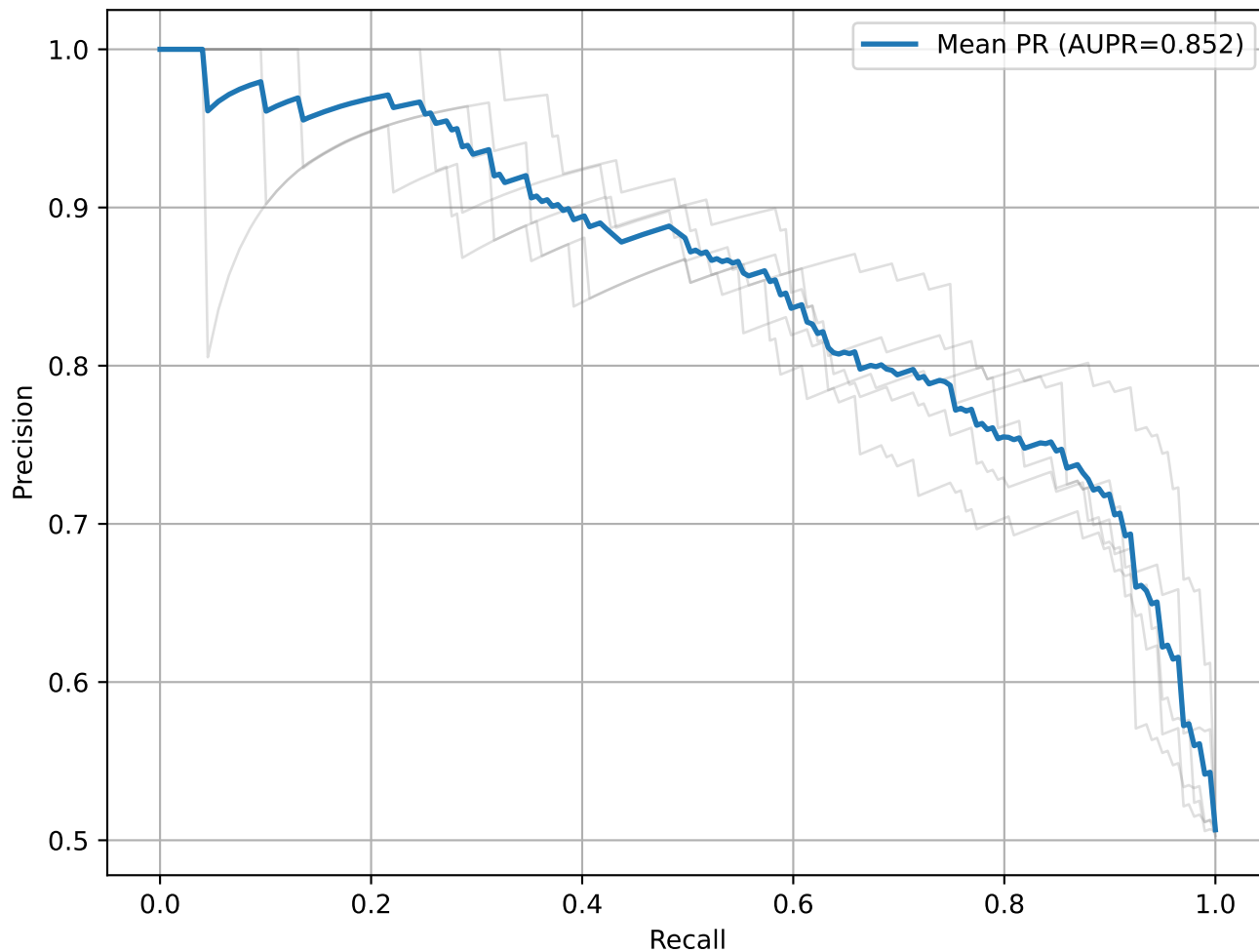
Classification Report - SVM (last fold)

	precision	recall	f1-score	support
0	0.82	0.75	0.78	89
1	0.78	0.84	0.81	92
accuracy			0.80	181
macro avg	0.80	0.79	0.79	181
weighted avg	0.80	0.80	0.80	181

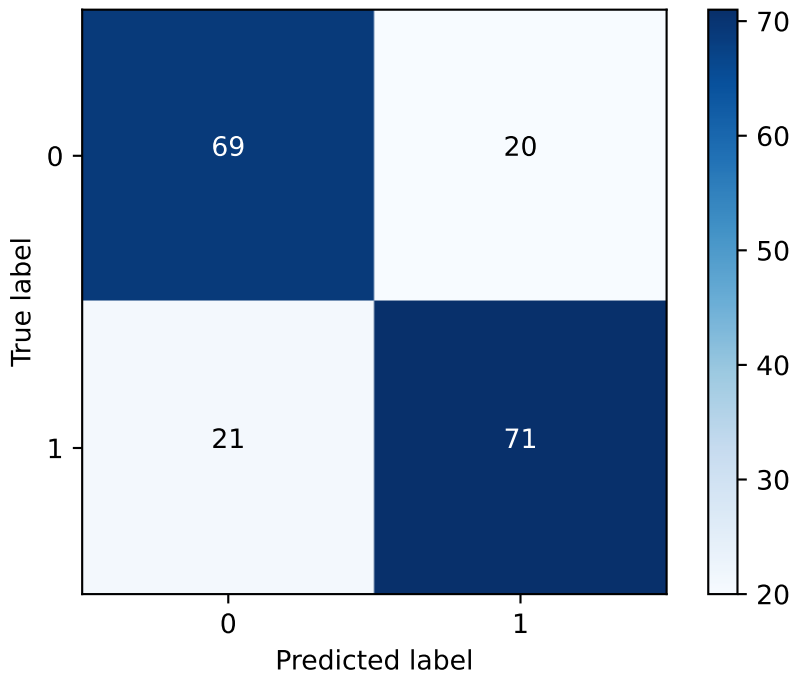
ROC - XGB



Precision-Recall - XGB



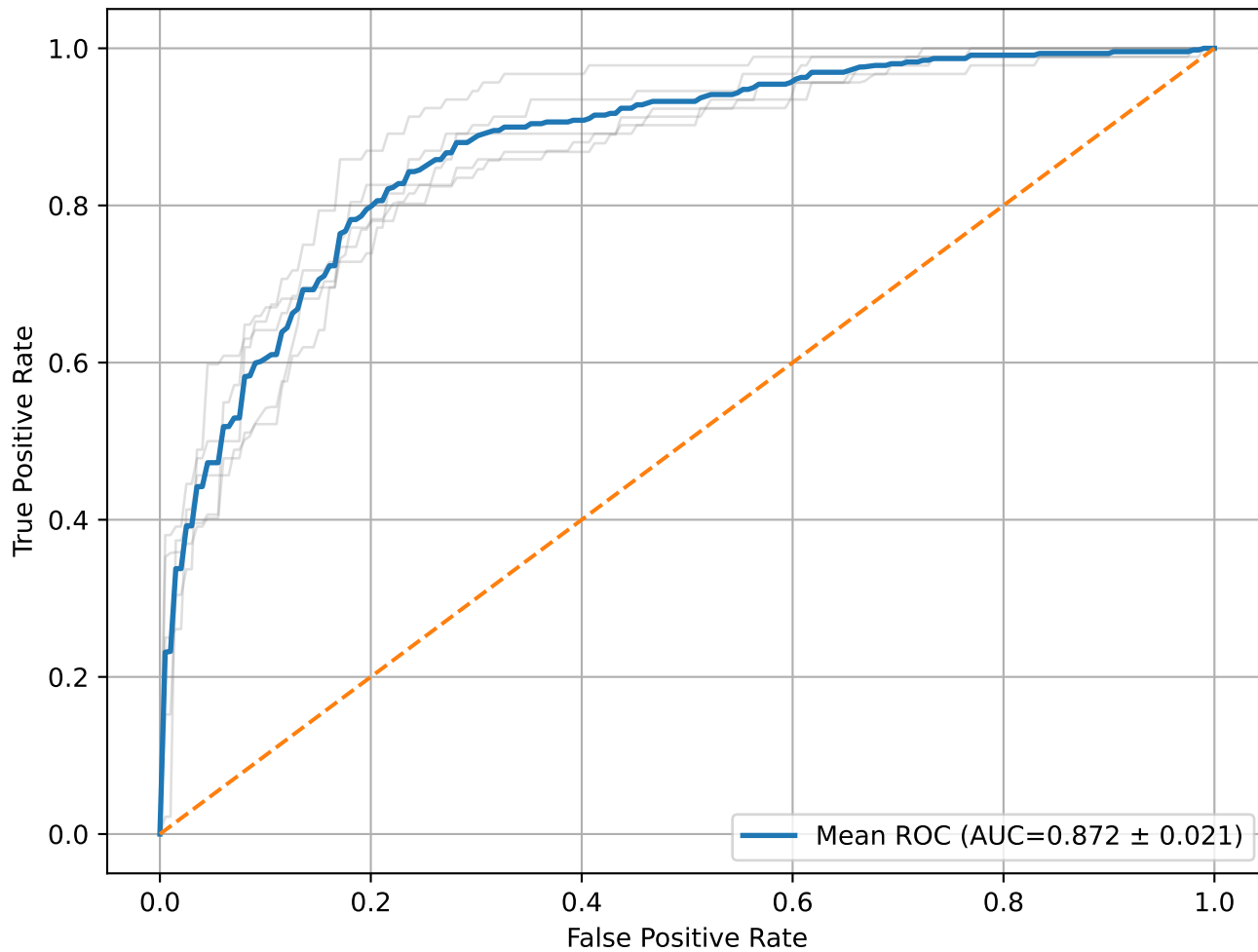
Confusion Matrix - XGB (last fold)



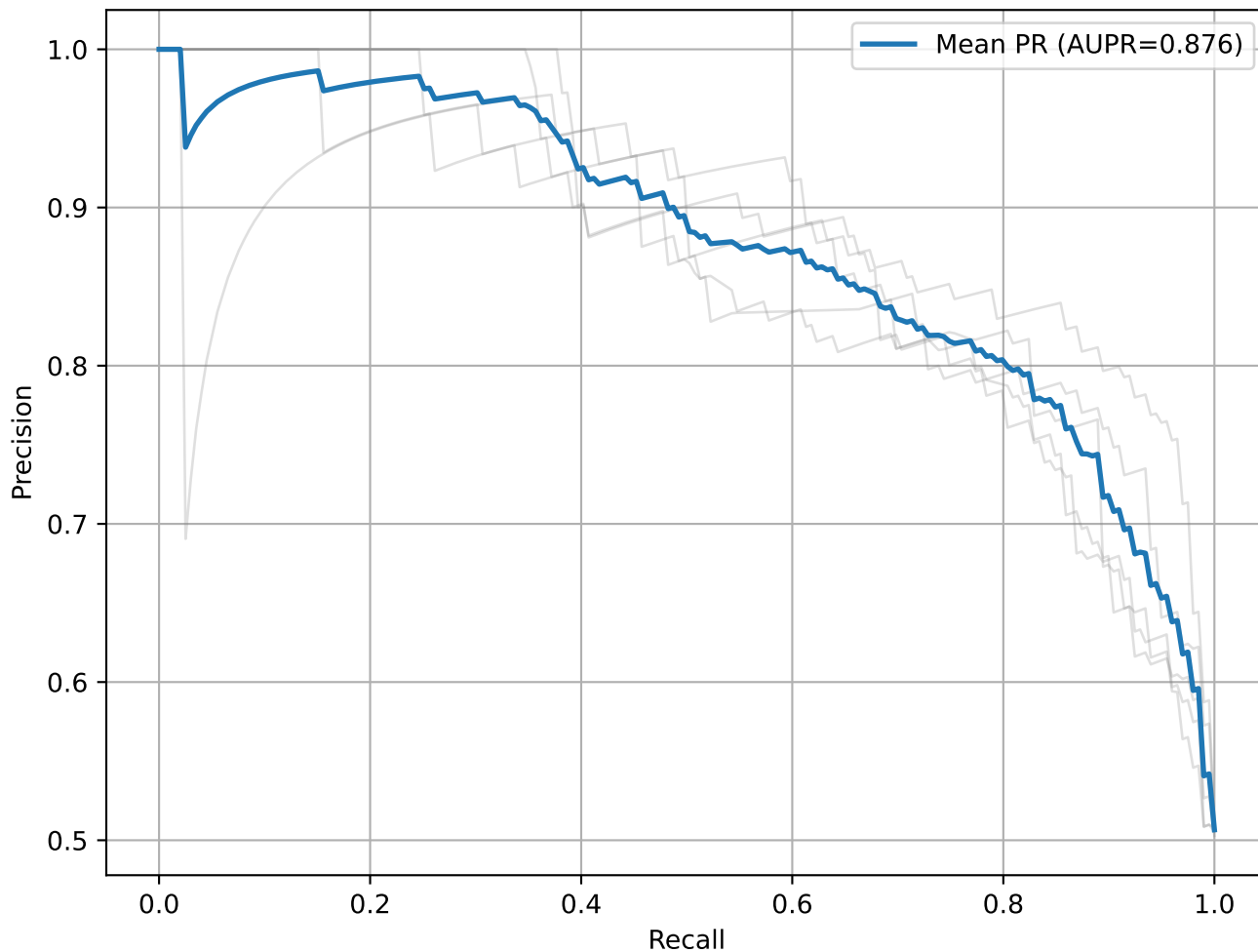
Classification Report - XGB (last fold)

	precision	recall	f1-score	support
0	0.77	0.78	0.77	89
1	0.78	0.77	0.78	92
accuracy			0.77	181
macro avg	0.77	0.77	0.77	181
weighted avg	0.77	0.77	0.77	181

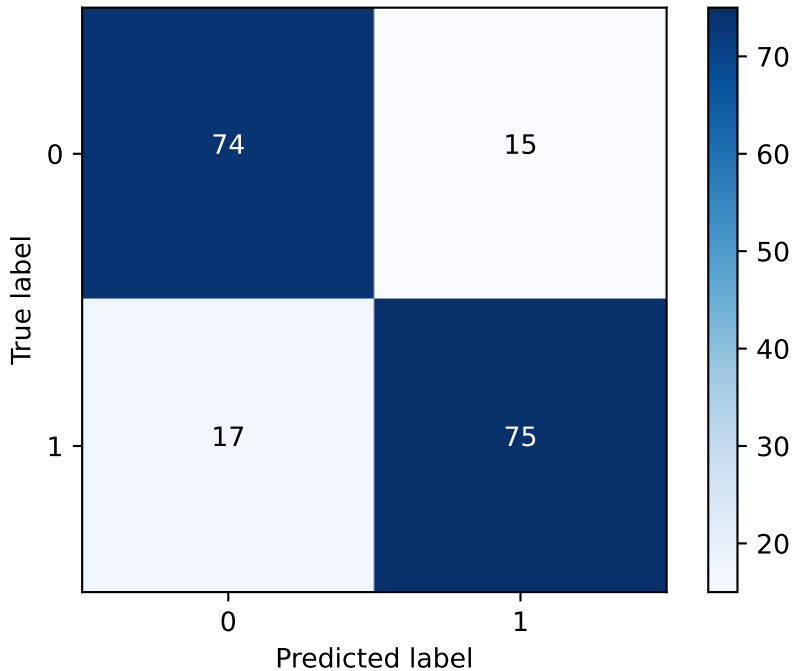
ROC - LGB



Precision-Recall - LGB



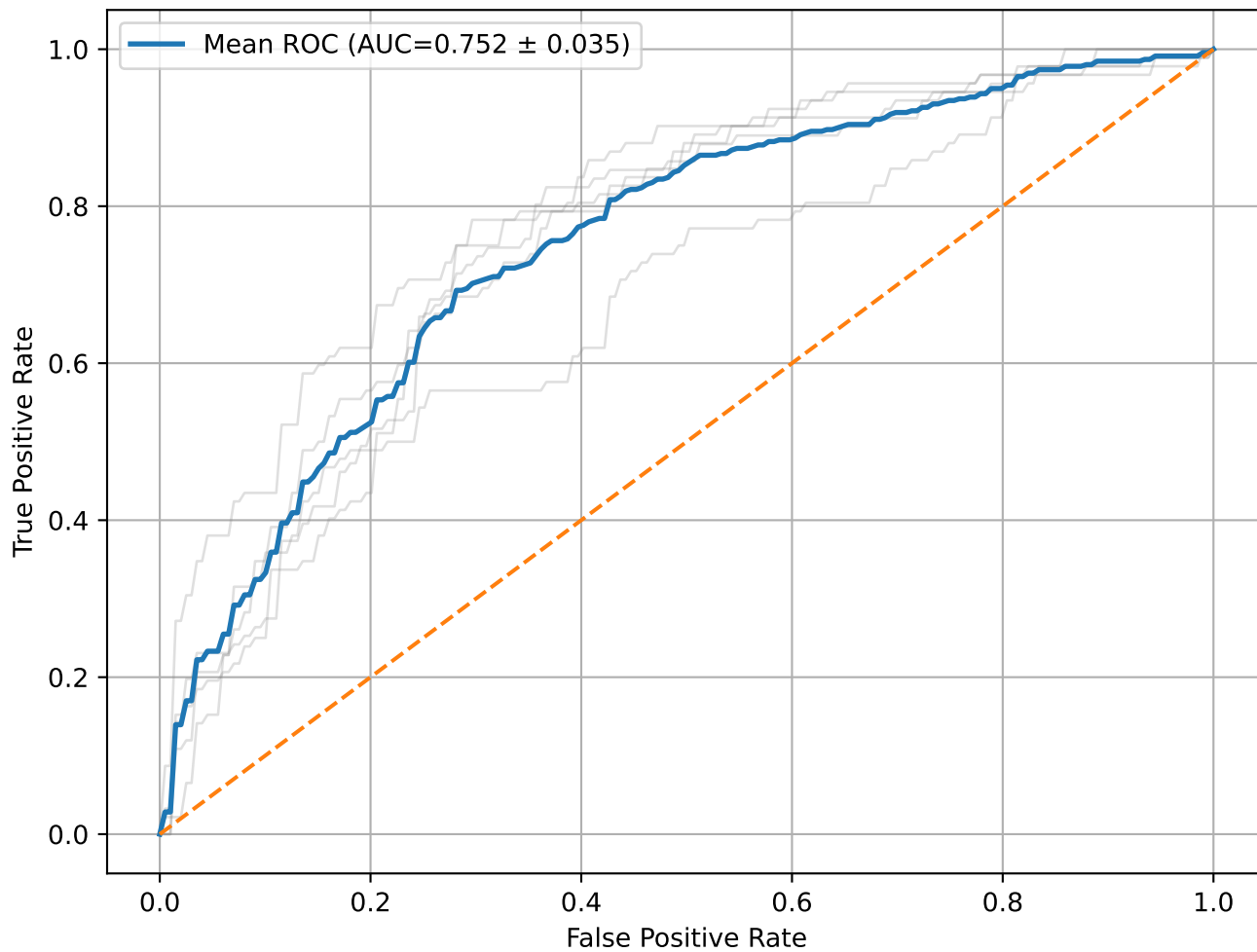
Confusion Matrix - LGB (last fold)



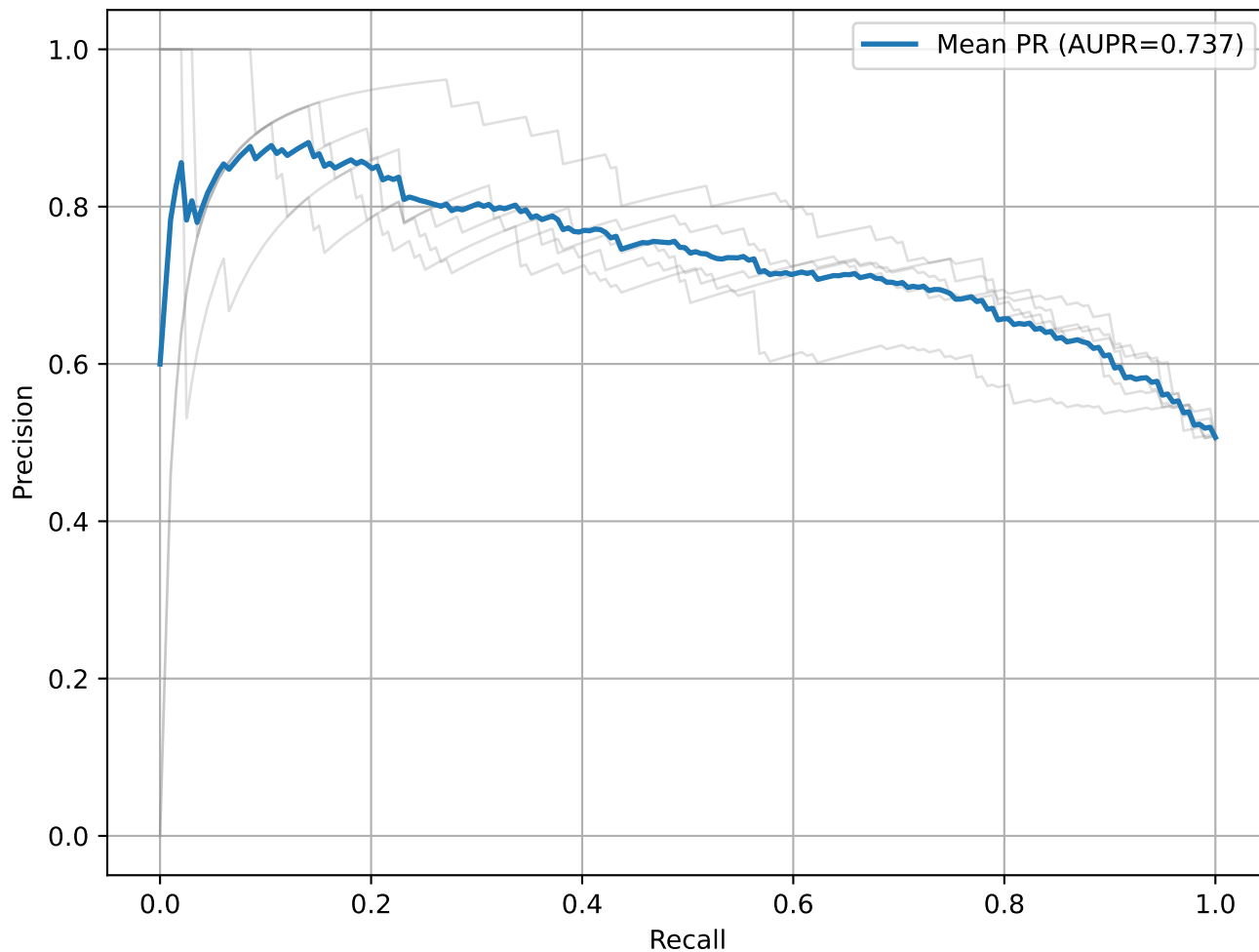
Classification Report - LGB (last fold)

	precision	recall	f1-score	support
0	0.81	0.83	0.82	89
1	0.83	0.82	0.82	92
accuracy			0.82	181
macro avg	0.82	0.82	0.82	181
weighted avg	0.82	0.82	0.82	181

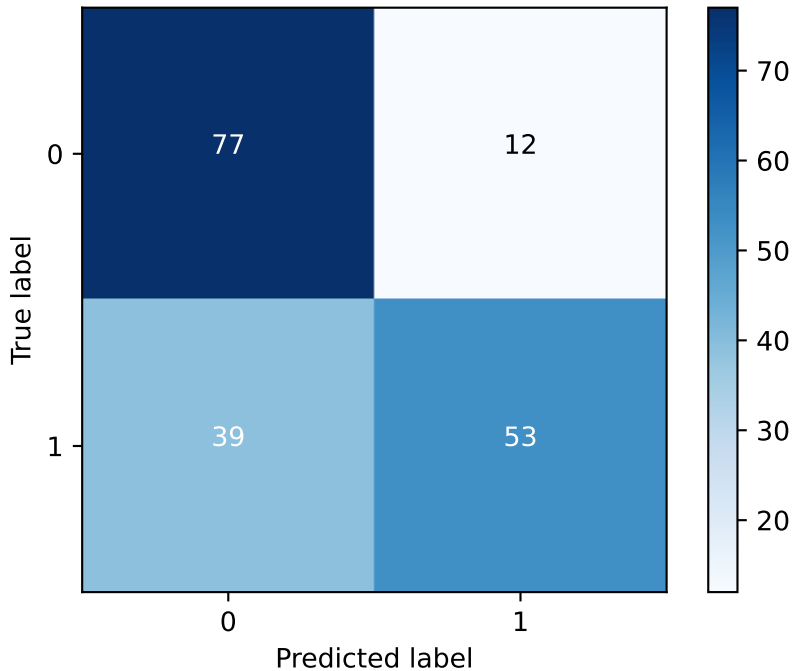
ROC - DNN



Precision-Recall - DNN



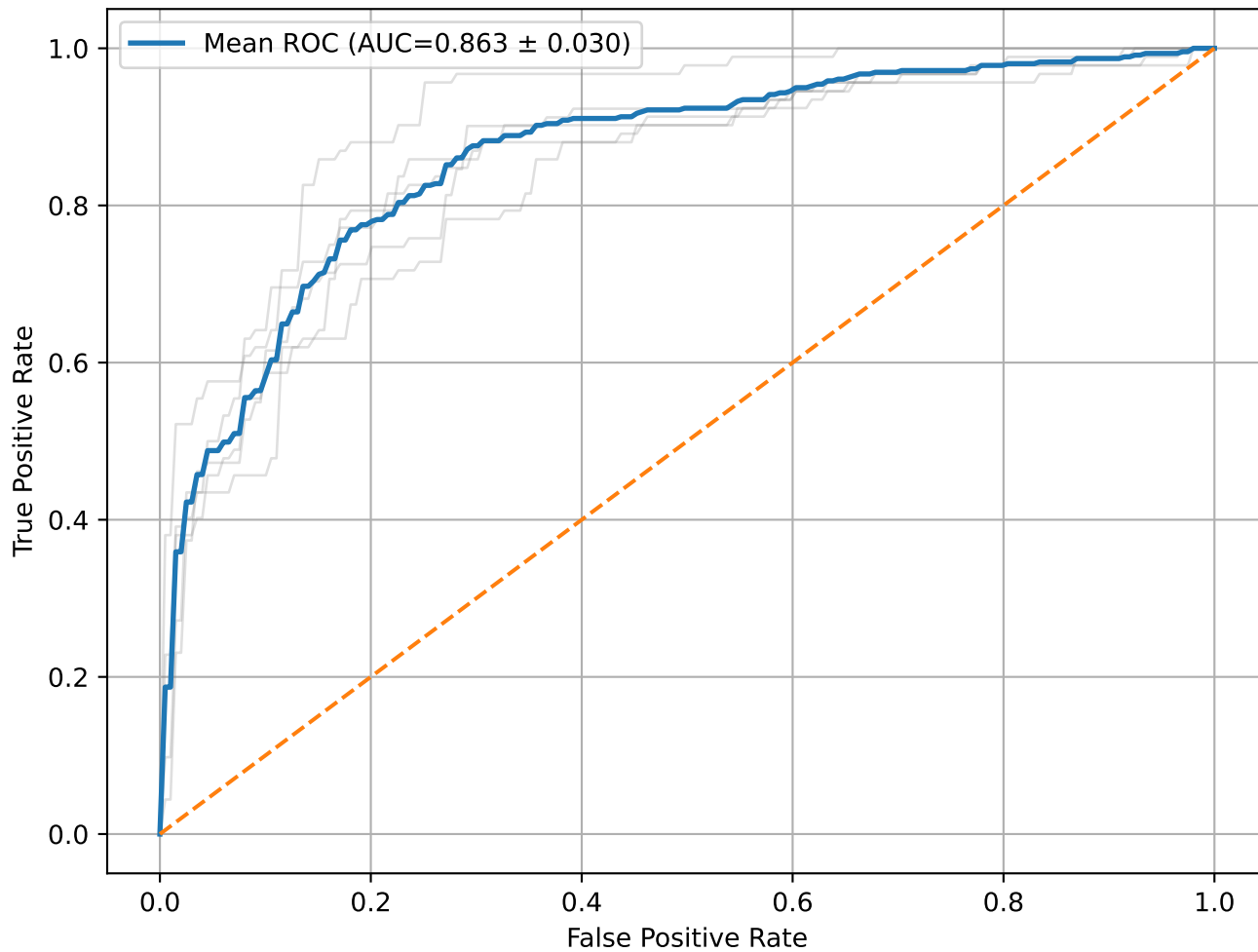
Confusion Matrix - DNN (last fold)



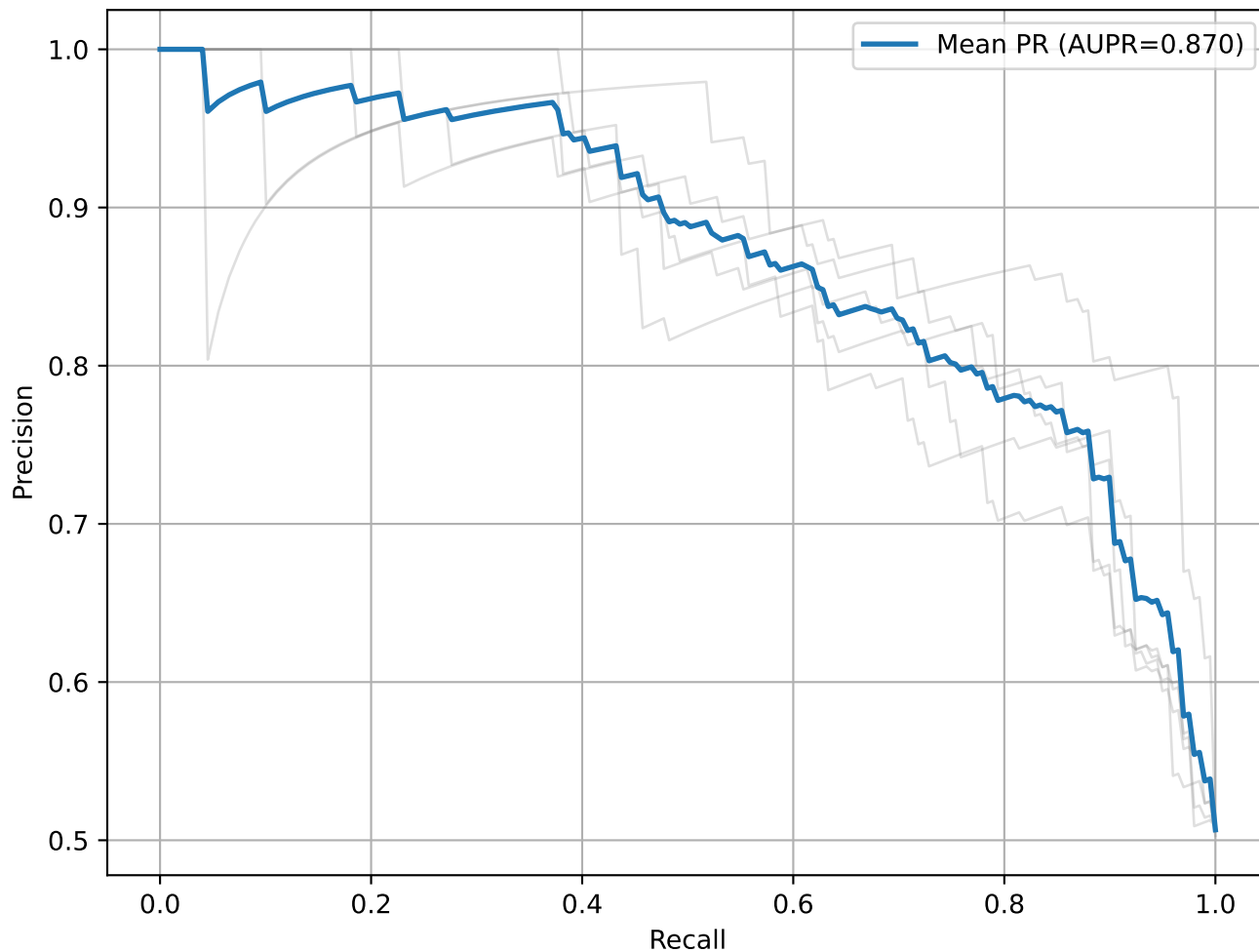
Classification Report - DNN (last fold)

	precision	recall	f1-score	support
0	0.66	0.87	0.75	89
1	0.82	0.58	0.68	92
accuracy			0.72	181
macro avg	0.74	0.72	0.71	181
weighted avg	0.74	0.72	0.71	181

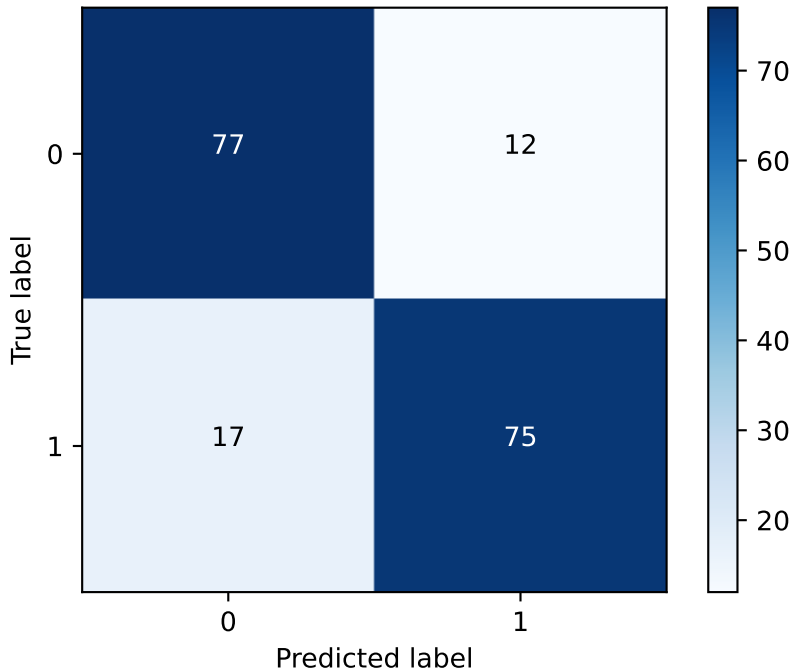
ROC - STACKING



Precision-Recall - STACKING



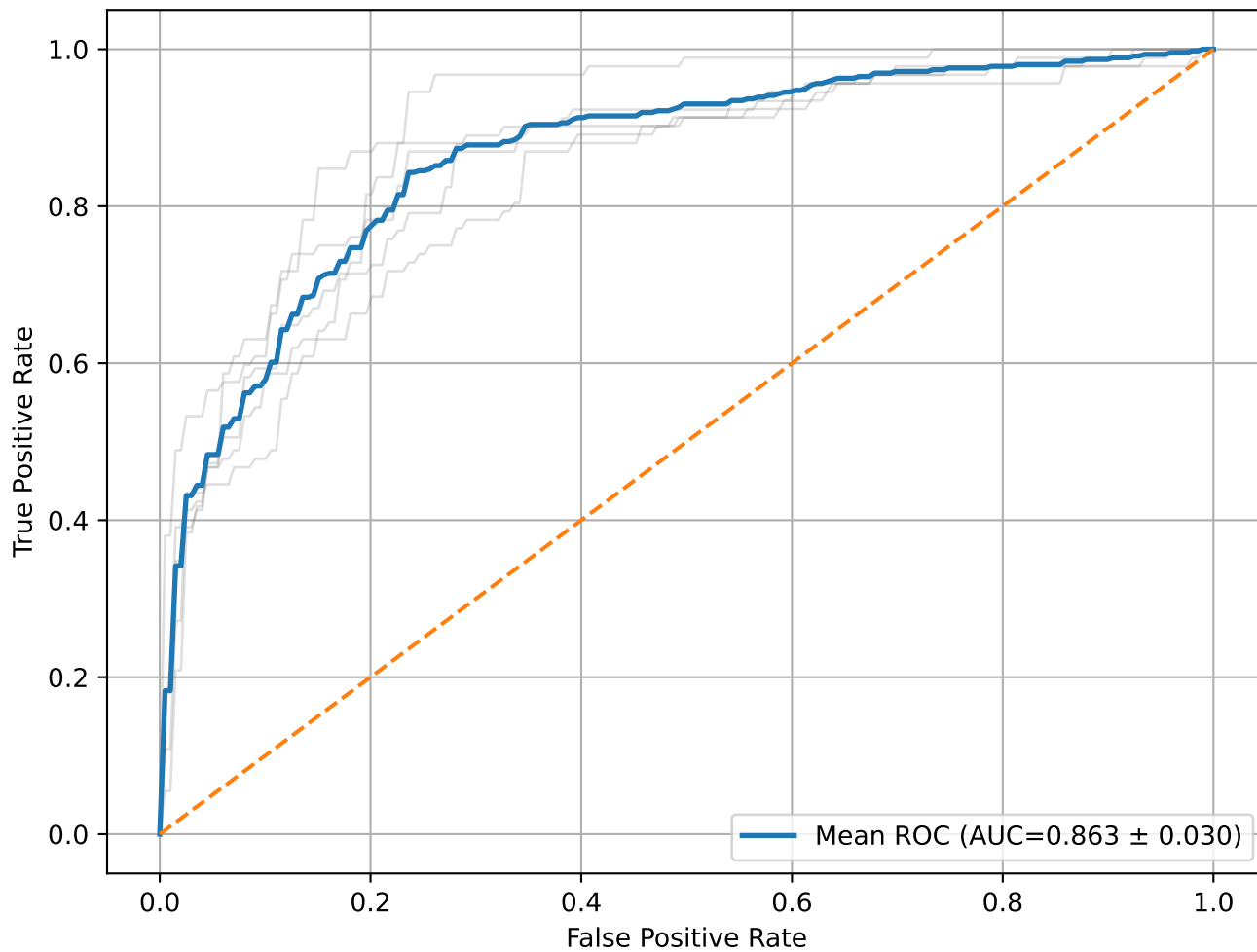
Confusion Matrix - STACKING (last fold)



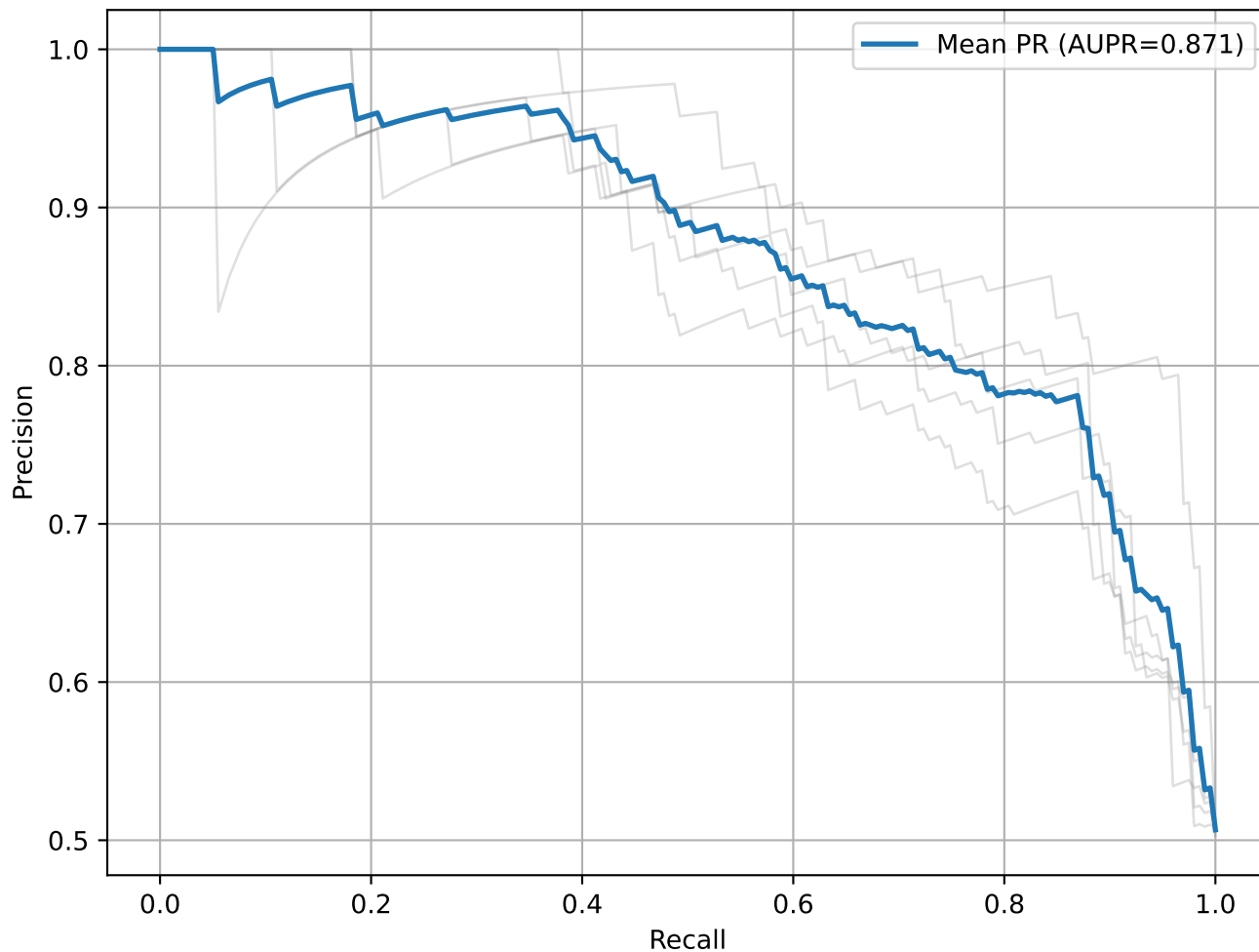
Classification Report - STACKING (last fold)

	precision	recall	f1-score	support
0	0.82	0.87	0.84	89
1	0.86	0.82	0.84	92
accuracy			0.84	181
macro avg	0.84	0.84	0.84	181
weighted avg	0.84	0.84	0.84	181

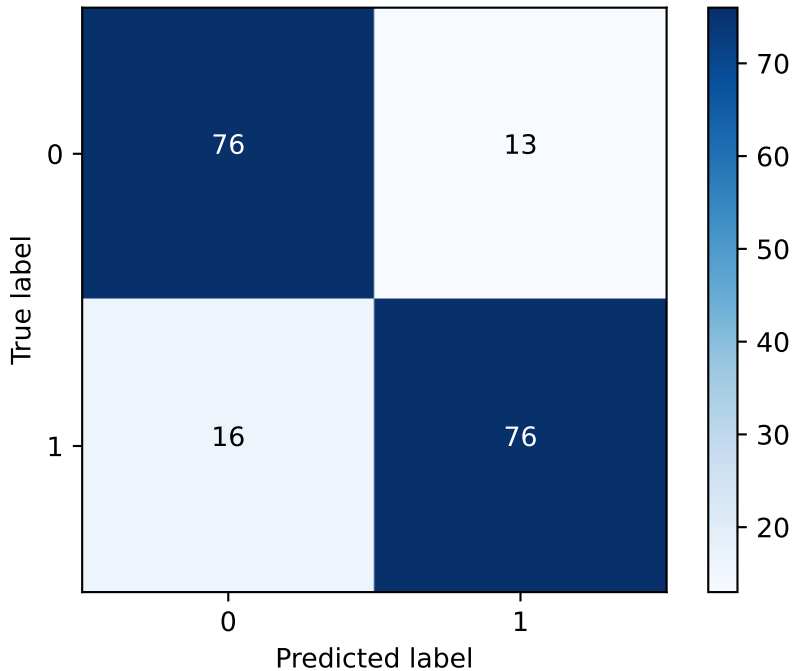
ROC - VOTING



Precision-Recall - VOTING



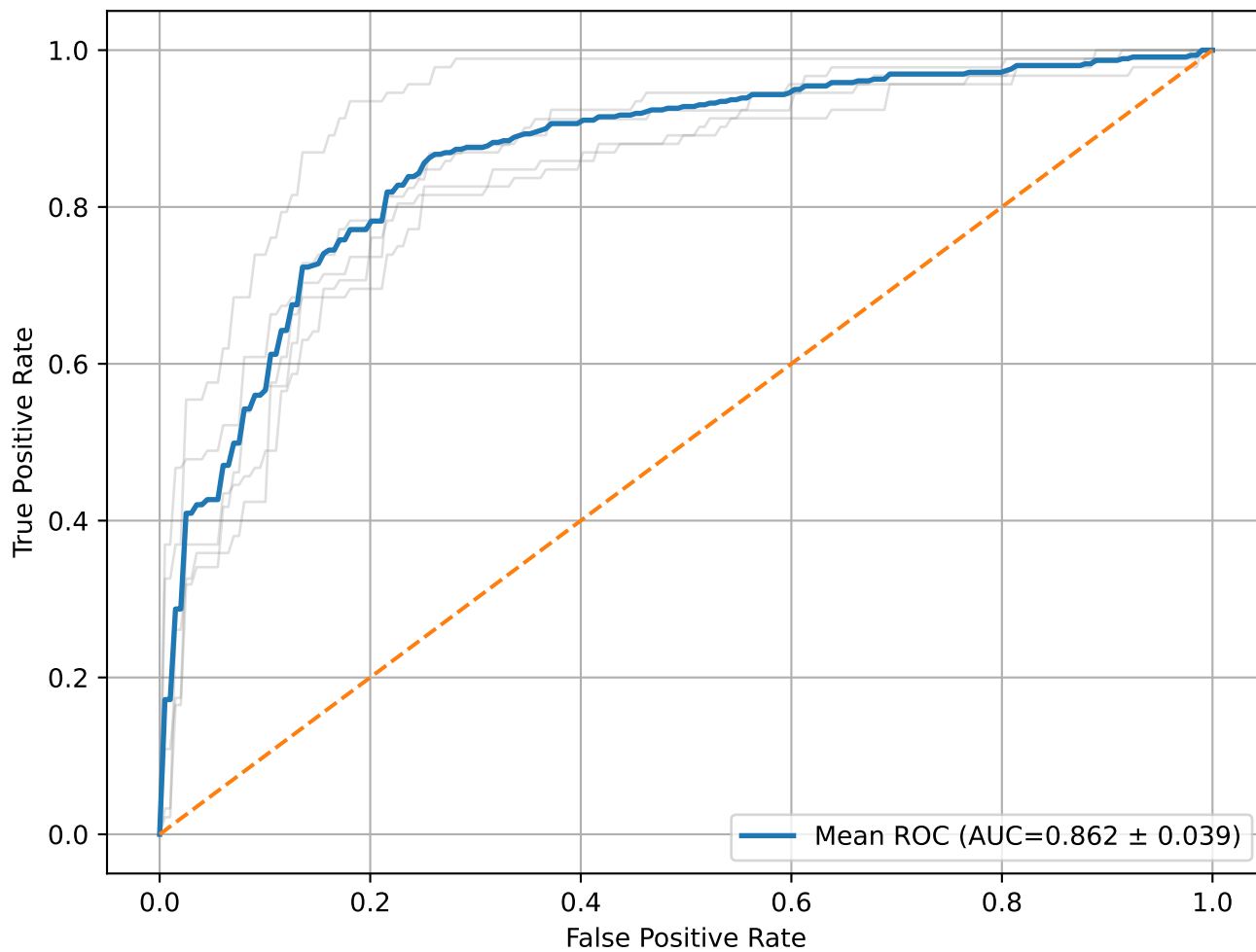
Confusion Matrix - VOTING (last fold)



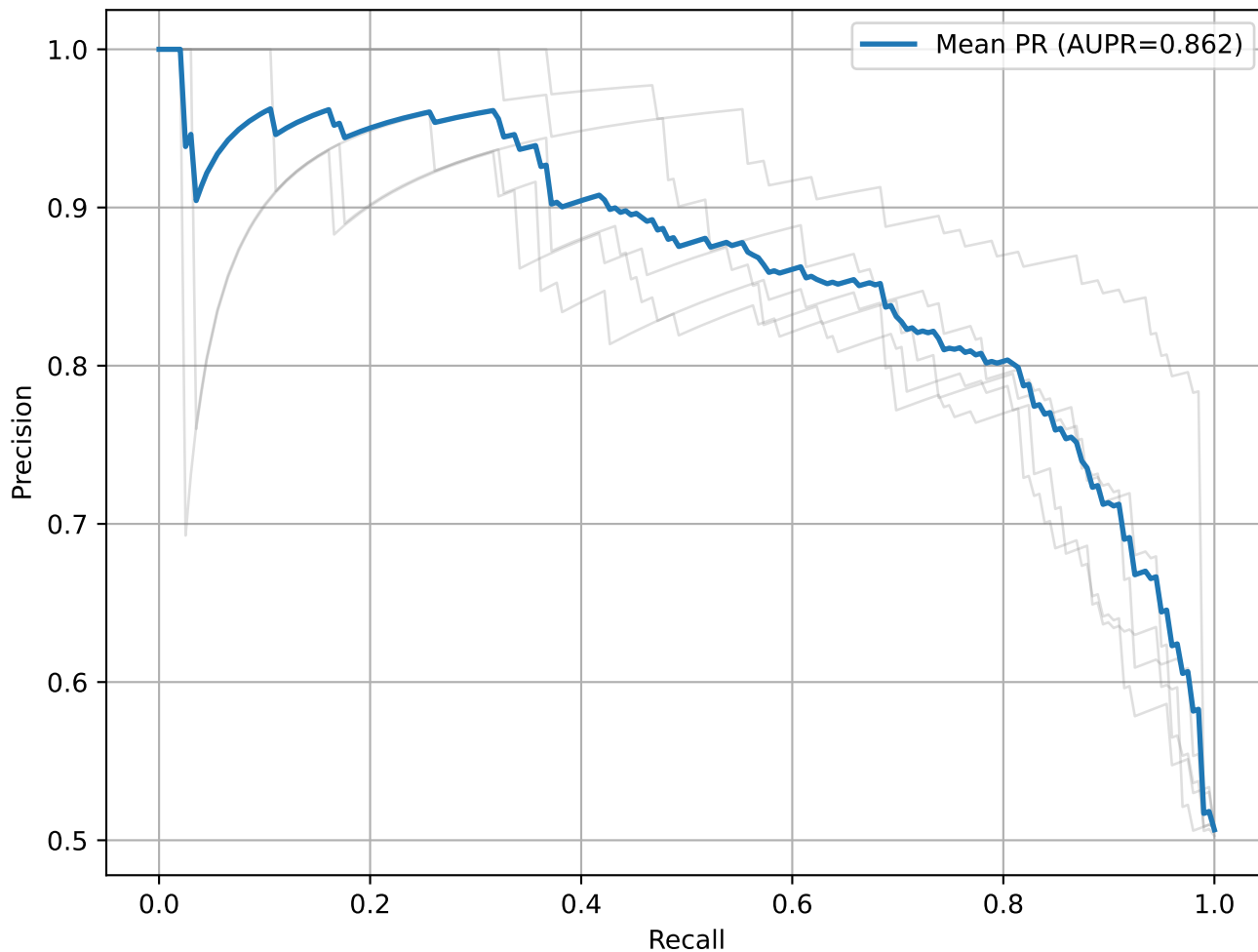
Classification Report - VOTING (last fold)

	precision	recall	f1-score	support
0	0.83	0.85	0.84	89
1	0.85	0.83	0.84	92
accuracy			0.84	181
macro avg	0.84	0.84	0.84	181
weighted avg	0.84	0.84	0.84	181

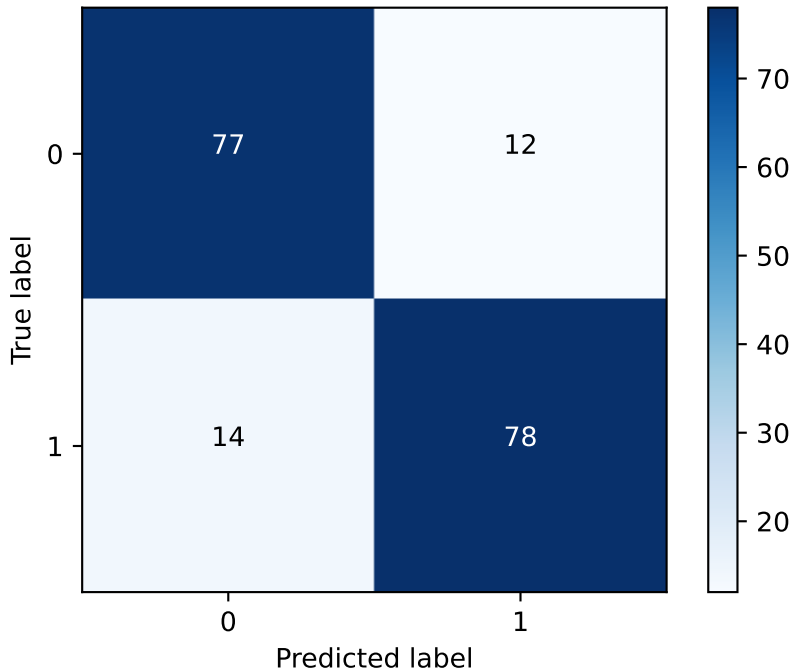
ROC - BAGGING



Precision-Recall - BAGGING



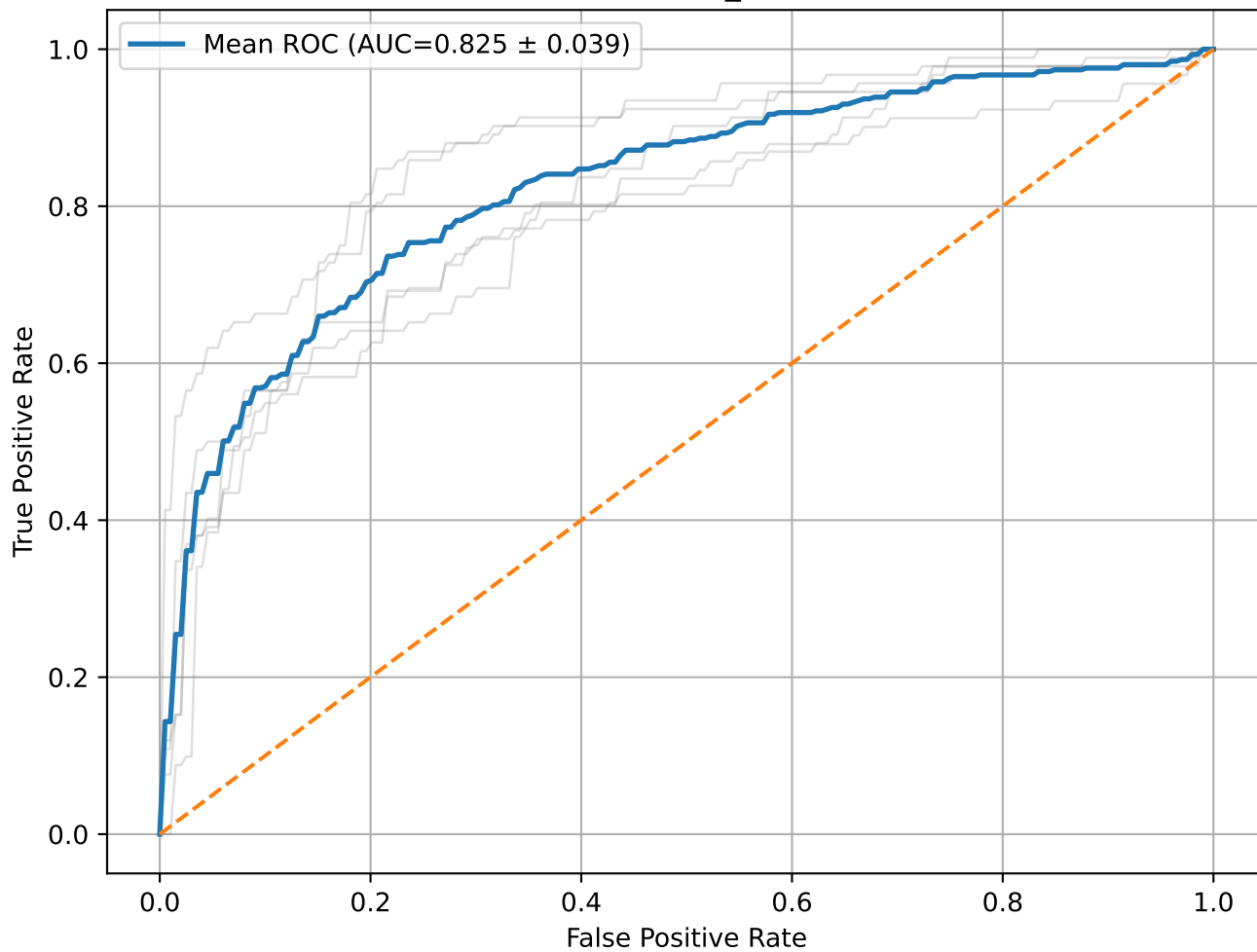
Confusion Matrix - BAGGING (last fold)



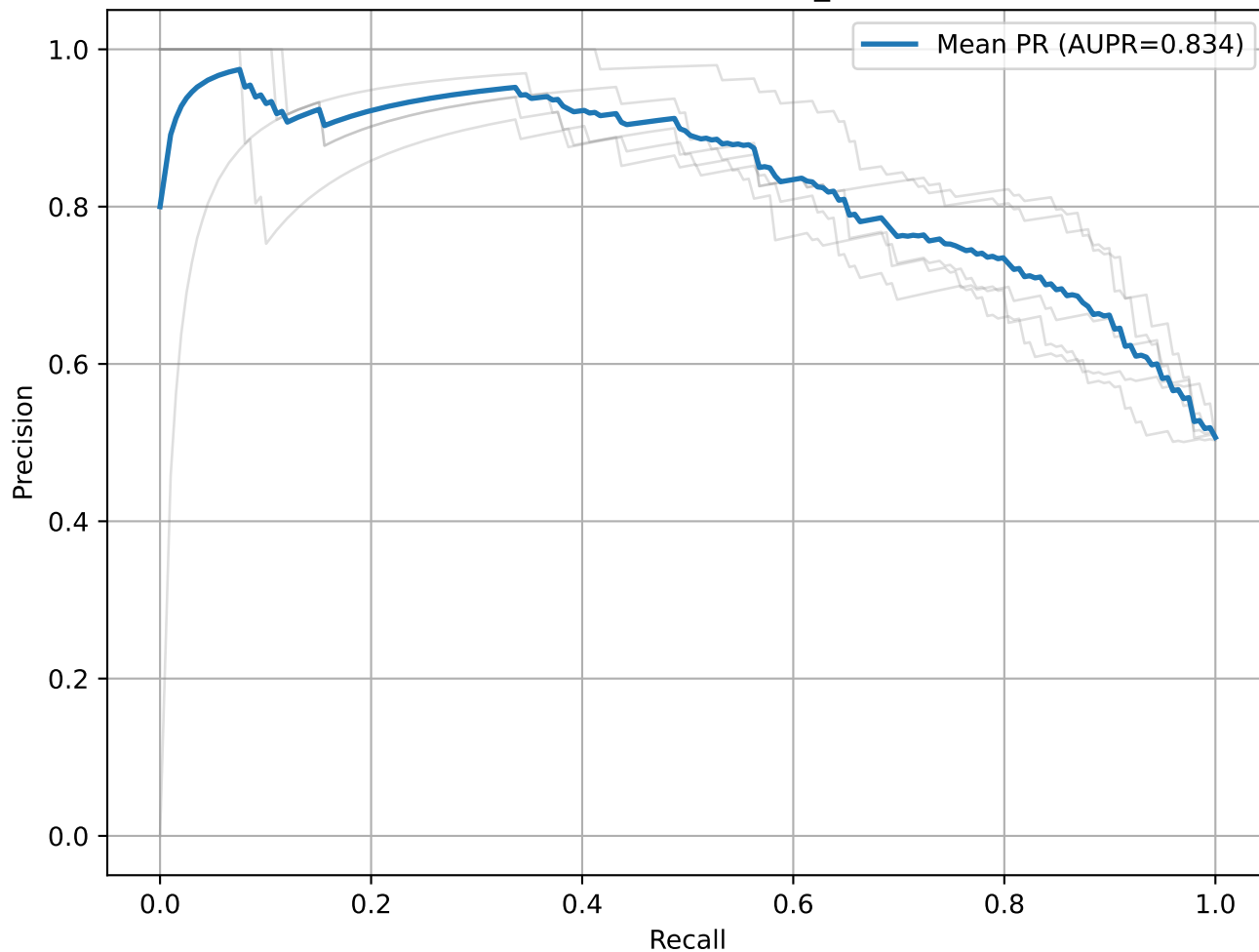
Classification Report - BAGGING (last fold)

	precision	recall	f1-score	support
0	0.85	0.87	0.86	89
1	0.87	0.85	0.86	92
accuracy			0.86	181
macro avg	0.86	0.86	0.86	181
weighted avg	0.86	0.86	0.86	181

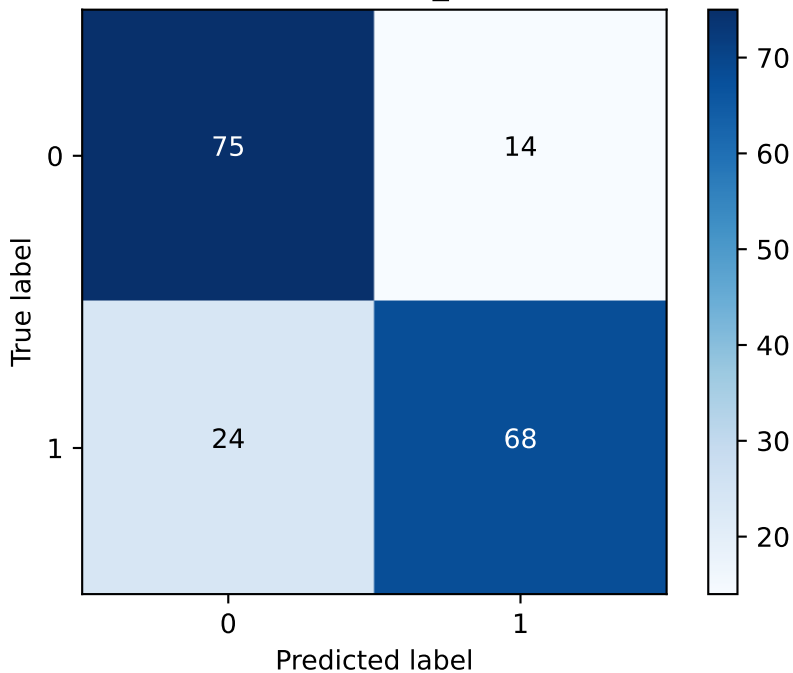
ROC - PCA_SVM



Precision-Recall - PCA_SVM



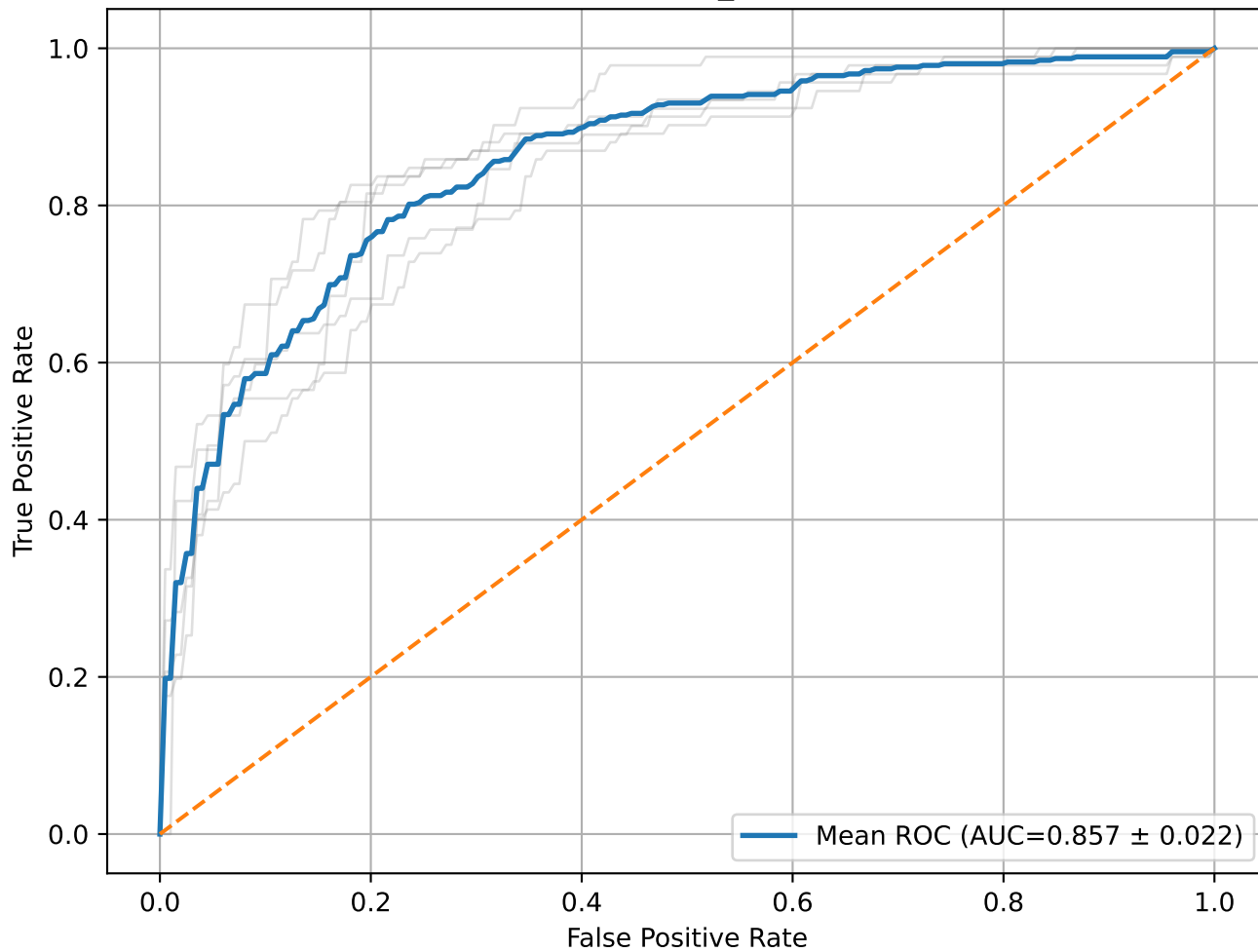
Confusion Matrix - PCA_SVM (last fold)



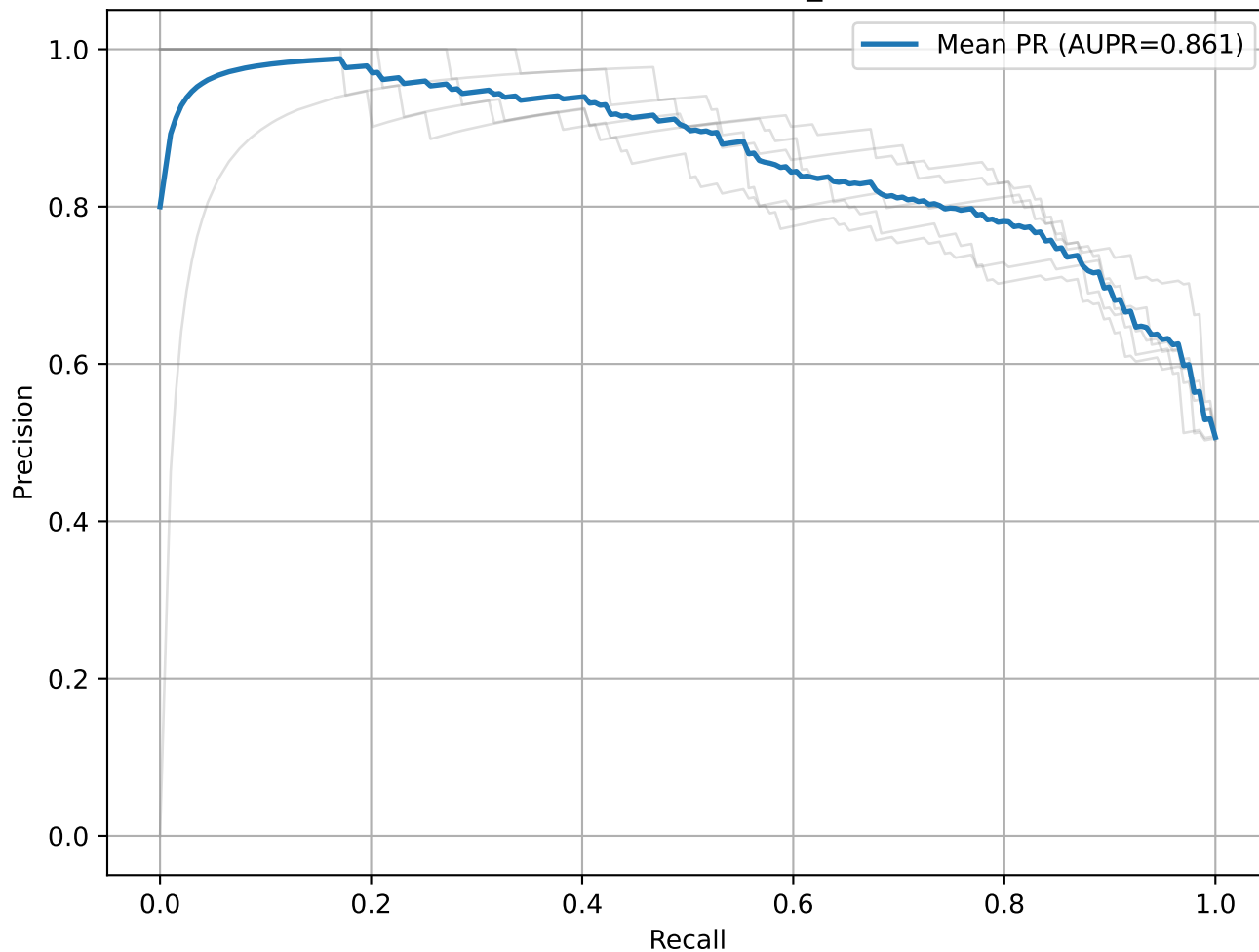
Classification Report - PCA_SVM (last fold)

	precision	recall	f1-score	support
0	0.76	0.84	0.80	89
1	0.83	0.74	0.78	92
accuracy			0.79	181
macro avg	0.79	0.79	0.79	181
weighted avg	0.79	0.79	0.79	181

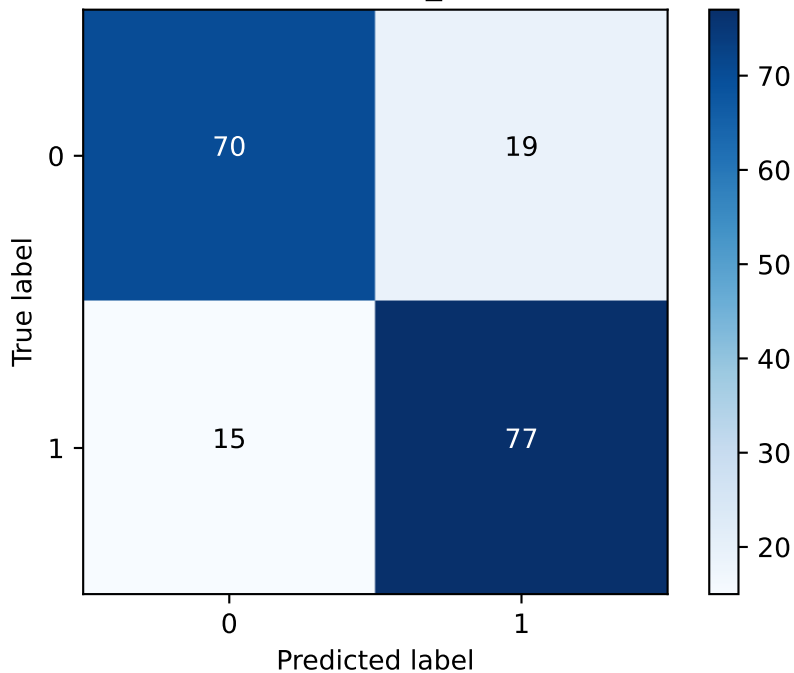
ROC - GA_XGB



Precision-Recall - GA_XGB



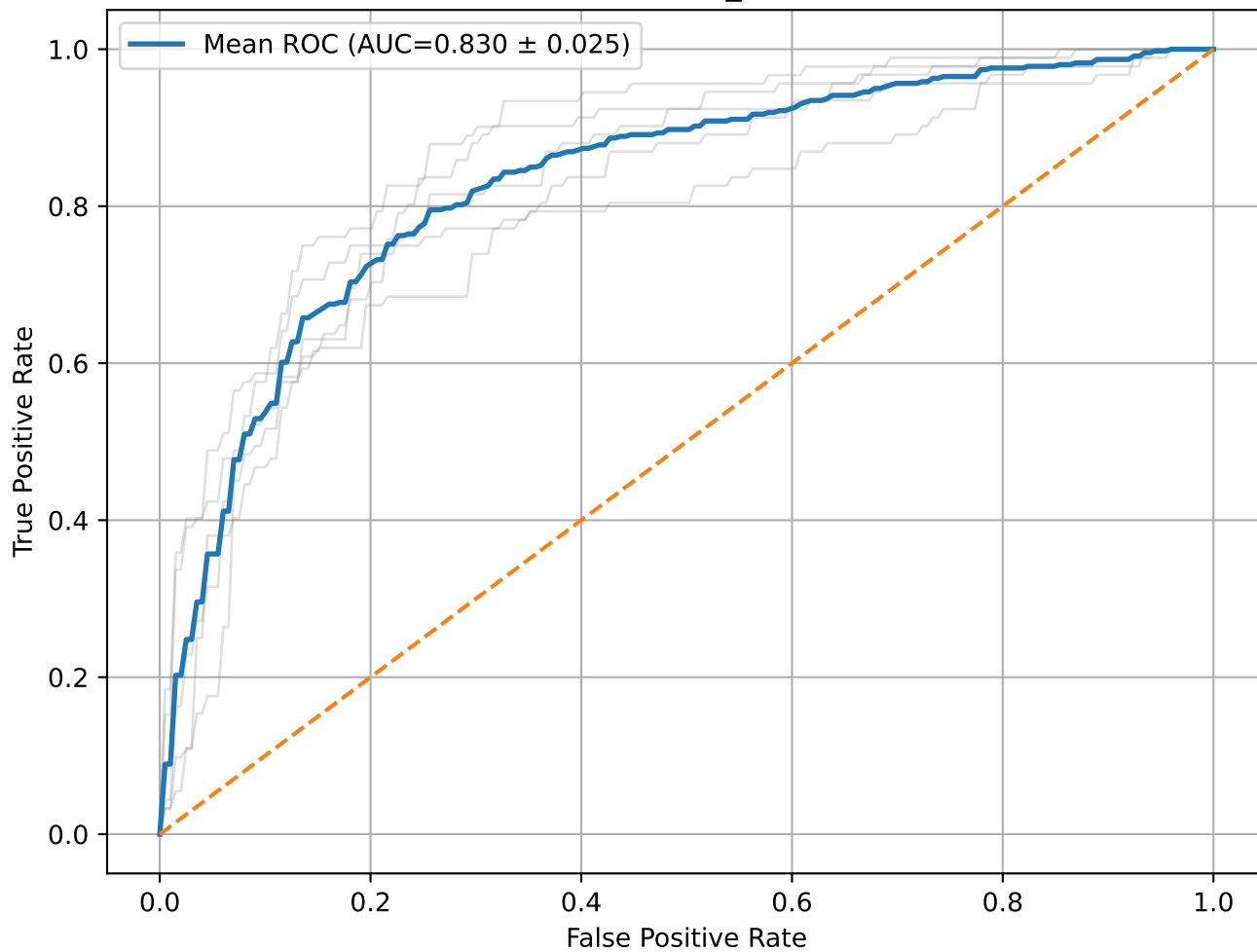
Confusion Matrix - GA_XGB (last fold)



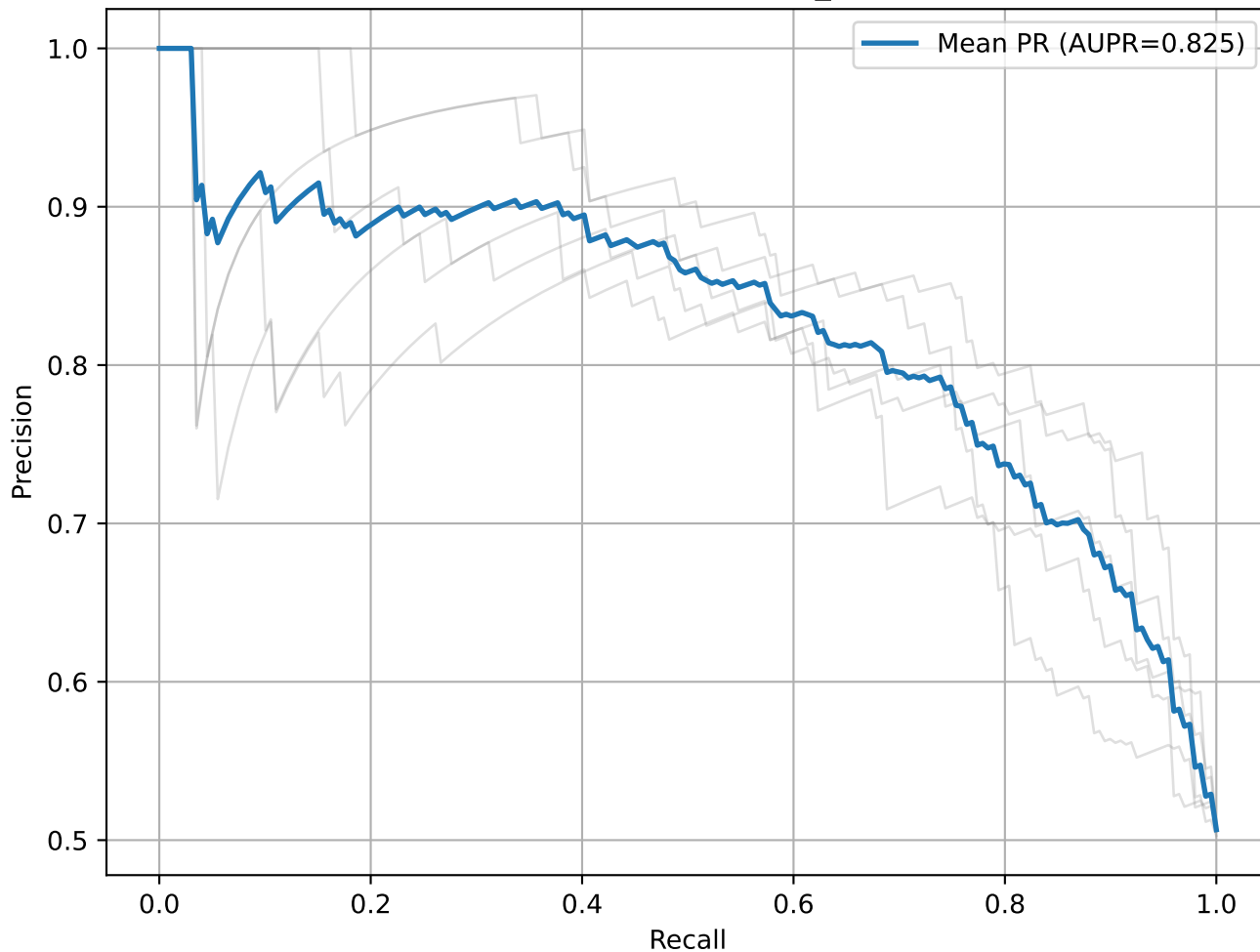
Classification Report - GA_XGB (last fold)

	precision	recall	f1-score	support
0	0.82	0.79	0.80	89
1	0.80	0.84	0.82	92
accuracy			0.81	181
macro avg	0.81	0.81	0.81	181
weighted avg	0.81	0.81	0.81	181

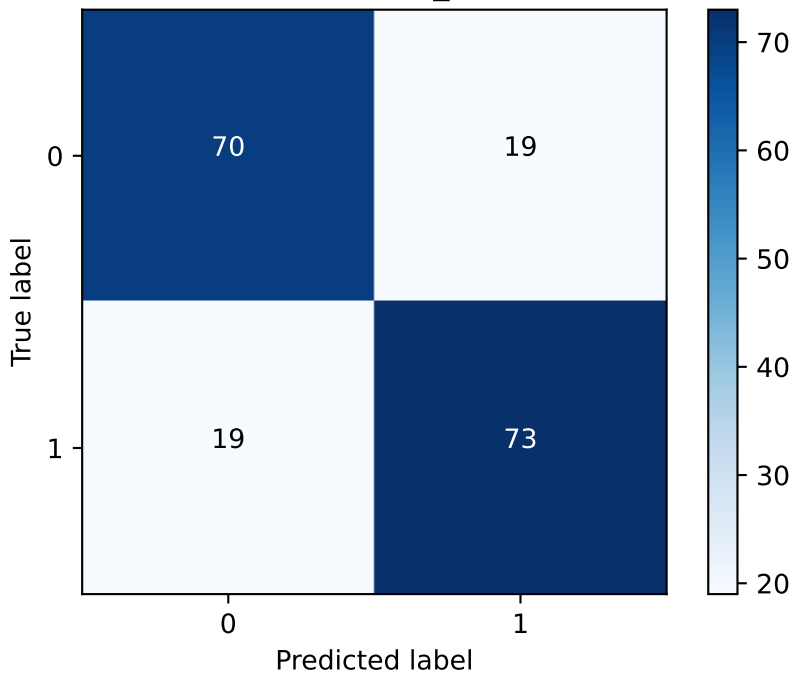
ROC - PSO_SVM



Precision-Recall - PSO_SVM



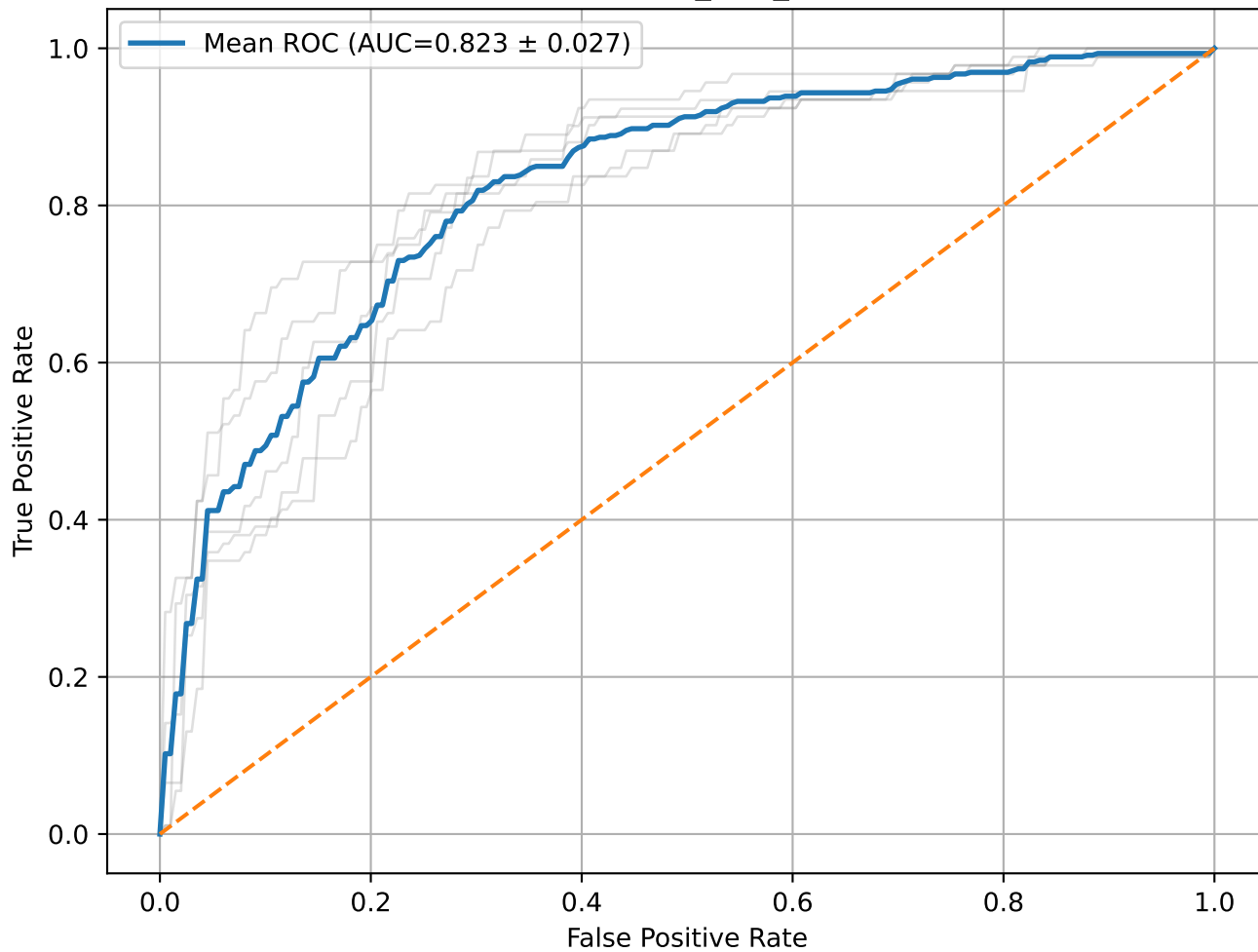
Confusion Matrix - PSO_SVM (last fold)



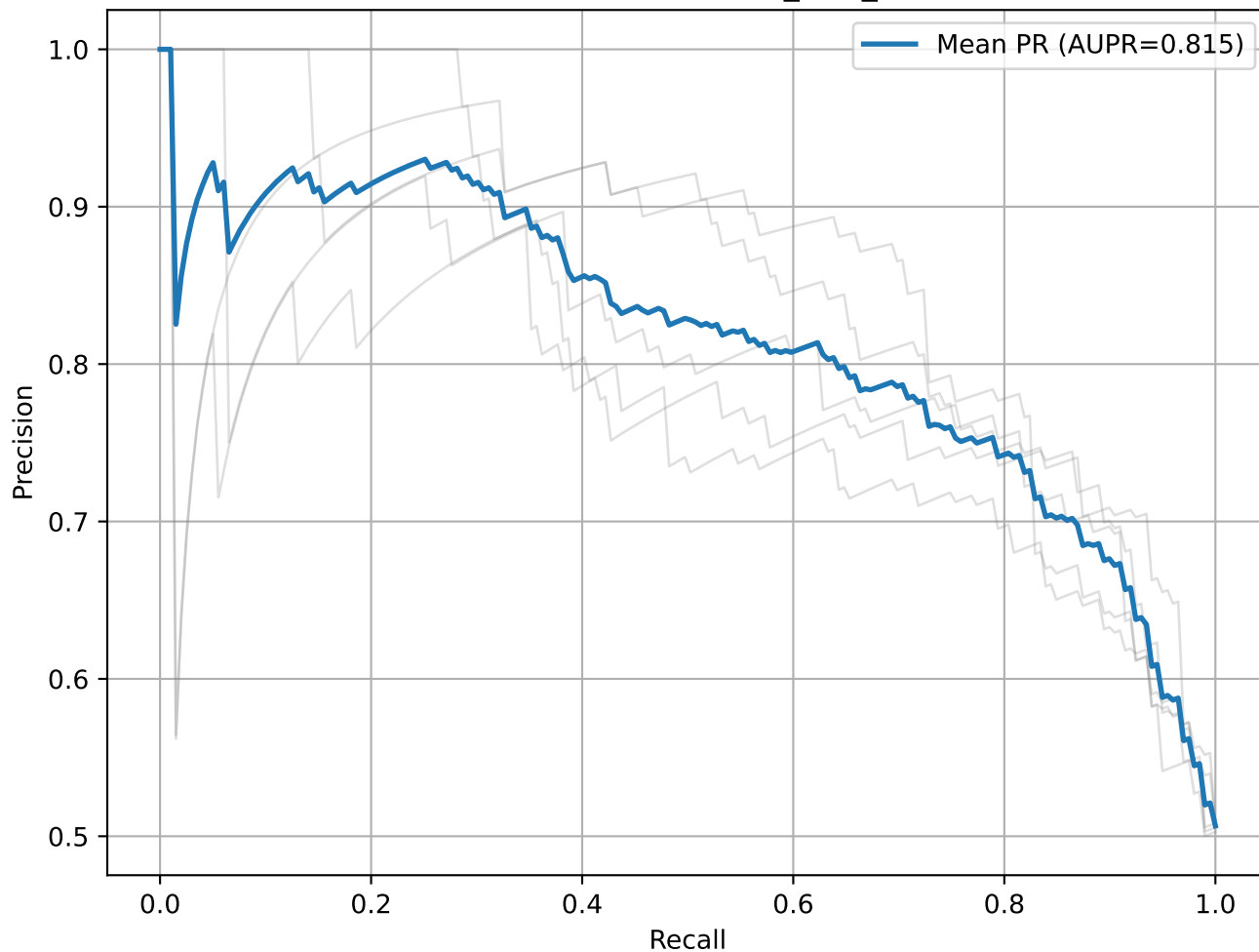
Classification Report - PSO_SVM (last fold)

	precision	recall	f1-score	support
0	0.79	0.79	0.79	89
1	0.79	0.79	0.79	92
accuracy			0.79	181
macro avg	0.79	0.79	0.79	181
weighted avg	0.79	0.79	0.79	181

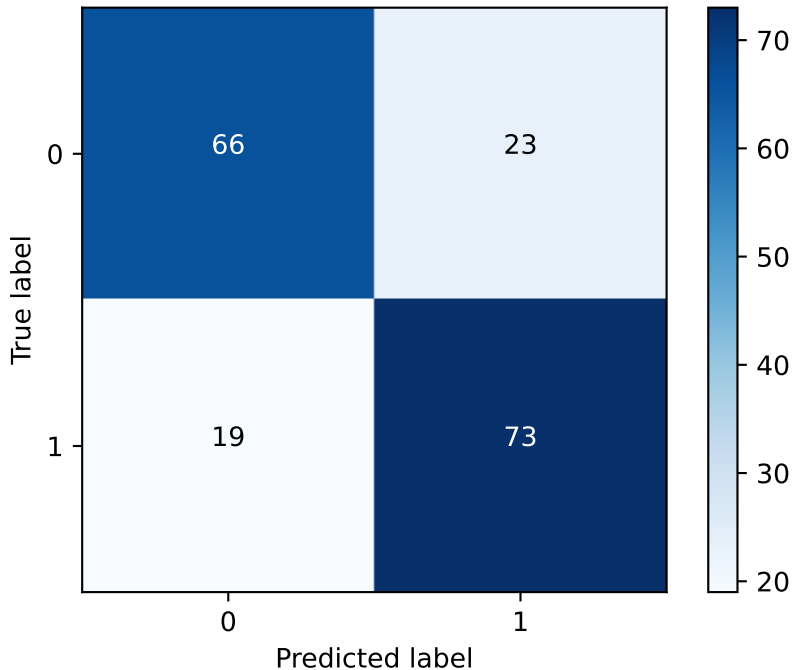
ROC - BAYES_OPT_SVMS



Precision-Recall - BAYES_OPT_SVMS



Confusion Matrix - BAYES_OPT_SVMS (last fold)



Classification Report - BAYES_OPT_SVMS (last fold)

	precision	recall	f1-score	support
0	0.78	0.74	0.76	89
1	0.76	0.79	0.78	92
accuracy			0.77	181
macro avg	0.77	0.77	0.77	181
weighted avg	0.77	0.77	0.77	181

Model Comparison Table (sorted by mean ROC AUC)

model	mean_roc_auc	std_roc_auc	mean_aupr	mean_accuracy	mean_f1
lgb	0.872859	0.021179	0.875835	0.798021	0.797520
voting	0.864235	0.029625	0.870645	0.789230	0.789081
stacking	0.863208	0.029962	0.870132	0.783723	0.783497
bagging	0.862754	0.038675	0.862161	0.800249	0.803993
ga_xgb	0.857954	0.021540	0.861407	0.784810	0.785663
xgb	0.849414	0.021392	0.852003	0.762747	0.763114
svm	0.836749	0.031514	0.823680	0.787007	0.792239
rf	0.836034	0.028549	0.850016	0.749505	0.748738
pso_svm	0.830594	0.025463	0.824647	0.760482	0.755918
pca_svm	0.825427	0.038847	0.834151	0.752826	0.750623
bayes_opt_svms	0.822990	0.026572	0.814883	0.750586	0.752970
logistic	0.819728	0.022263	0.809579	0.756099	0.757415
dnn	0.751860	0.034504	0.736736	0.689891	0.666518

Combined ROC - All Models

